

Chapter 3

Linear Series

Revised chapter corresponding to original pp. 158–220.

Original numbering and proof order retained.

The accompanying correction report gives the original page locations.

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In this chapter all schemes will be considered over $\mathbb{C}$ unless another ground field is explicitly specified. Schemes are of finite type over the ground field unless stated otherwise. Some results hold over more general fields; characteristic-zero assumptions are retained where they are needed.

3.1 Asymptotic Theory

3.1A Basic Definitions

“This definition may look a little bit scary.”

Caucher Birkar on the asymptotic definition of Kodaira dimension

The asymptotic behavior of the linear series $|L^{\otimes m}|$ as $m \rightarrow \infty$, due to Iitaka and Ueno, is that up to birational equivalence they define an essentially unique rational mapping $\varphi : X \dashrightarrow Y$, and L is close to trivial (in a suitable sense) along the fibres of φ . Throughout this section X always means a projective variety.

Definition 3.1.1 (Semigroup and exponent of a line bundle). Let L be a line bundle on a projective variety X . The semigroup of L consists of those non-negative powers of L that have a non-zero section:

$$N(L) = N(X, L) = \{m \geq 0 \mid H^0(X, L^{\otimes m}) \neq 0\}.$$

Clearly $N(L) = \{0\}$ if $H^0(X, L^{\otimes m}) = 0$ for all $m > 0$. Assuming that $N(L) \neq \{0\}$, let $e = e(L)$ be the greatest common divisor of $\{m \in N(L) \mid m > 0\}$, called the *exponent* of L . Every element of $N(L)$ is a multiple of e , and conversely every sufficiently large multiple of e belongs to $N(L)$. The semigroup $N(X, D)$ and exponent $e(D)$ of a Cartier divisor D are defined via $L = \mathcal{O}_X(D)$.

For completeness, the last assertion is an elementary fact about semigroups. After dividing by e , choose finitely many elements with greatest common divisor 1, and choose a positive element a . Their non-negative sums represent every residue class modulo a : the subsemigroup they generate in the finite group $\mathbb{Z}/a\mathbb{Z}$ is a subgroup, and is the whole group. Choose one representative b_r in the semigroup for each residue r . Every sufficiently large integer congruent to r is $b_r + ta$ with $t \geq 0$. Here “sufficiently divisible” means divisible by a fixed positive integer; when a lower bound is also needed, we say “sufficiently large and divisible.”

Example 3.1.2. Let T be a projective variety of dimension $d \geq 1$ carrying a non-trivial torsion line bundle L_η of order e in $\text{Pic}(T)$, so that $L_\eta^{\otimes e} \cong \mathcal{O}_T$. Let Y be any projective variety of dimension k , and B a very ample line bundle on Y . Set

$$X = Y \times T, \quad L = \text{pr}_1^* B \otimes \text{pr}_2^* L_\eta.$$

Then $e(L) = e$ and $N(L) = e\mathbb{Z}_{\geq 0}$, because the Künneth formula gives

$$H^0(X, L^{\otimes m}) \cong H^0(Y, B^{\otimes m}) \otimes H^0(T, L_\eta^{\otimes m}).$$

Here we use the fact that a non-trivial torsion line bundle $\mathcal{L}$ on a projective variety has no non-zero section. Indeed, if $s \neq 0$ were a section and $\mathcal{L}^{\otimes a} \cong \mathcal{O}_T$, then s^a would be a non-zero constant. Hence s would be nowhere vanishing and would trivialize $\mathcal{L}$, a contradiction.

For $m > 0$ in $N(L)$, consider the rational map

$$\varphi_m = \varphi_{|L^{\otimes m}|} : X \dashrightarrow \mathbb{P}H^0(X, L^{\otimes m})$$

associated to the complete linear series, and denote by $Y_m = \overline{\varphi_m(X)}$ the closure of its image. Here projective space uses the quotient convention, as in Chapter 1. Our purpose is to understand the asymptotic behavior of these rational maps as $m \rightarrow \infty$.

Definition 3.1.3 (Iitaka dimension). Assume that X is normal. The Iitaka dimension of L is

$$\kappa(L) = \kappa(X, L) = \max_{\substack{m \in N(L) \\ m > 0}} \dim \varphi_m(X),$$

provided that $N(L)$ is non-trivial; otherwise set $\kappa(L) = -\infty$. If X is non-normal, pass to its normalization $\nu : X' \rightarrow X$ and set

$$\kappa(X, L) = \kappa(X', \nu^*L).$$

The definition for Cartier divisors is similar.

Thus either $\kappa(X, L) = -\infty$, or $0 \leq \kappa(X, L) \leq \dim X$. With this convention, on a non-normal variety $\kappa(X, L)$ need not measure the growth of $H^0(X, L^{\otimes m})$ itself; see Example 3.1.7.

Example 3.1.4. For the pair (X, L) in Example 3.1.2 and a positive multiple m of e ,

$$H^0(X, L^{\otimes m}) \cong H^0(Y, B^{\otimes m}) \otimes \mathbb{C}.$$

The image $\varphi_m(X)$ is the image of Y under the embedding defined by $B^{\otimes m}$. Applying the same observation after normalization if necessary gives $\kappa(X, L) = \dim Y = k$.

Example 3.1.5 (Kodaira dimension). Let X be a smooth projective variety and K_X a canonical divisor. Then $\kappa(X) = \kappa(X, K_X)$ is the Kodaira dimension of X . For a singular variety, its Kodaira dimension is defined using any smooth projective model; this is independent of the model by birational invariance of the pluricanonical spaces.

Example 3.1.6 (Iitaka dimension of restrictions). Iitaka dimension can be unpredictable under restriction to subvarieties. Let $X = \text{Bl}_p \mathbb{P}^2$, with exceptional curve E , and let H be the pullback of a line. Both $\mathcal{O}_X(H)$ and $\mathcal{O}_X(H + E)$ have

Itaka dimension 2, but their restrictions to E have Itaka dimensions 0 and $-\infty$, respectively. Indeed, by the projection formula $(H \cdot E) = 0$, and

$$\mathcal{O}_X(H)|_E \cong \mathcal{O}_{\mathbb{P}^1}, \quad \mathcal{O}_X(H + E)|_E \cong \mathcal{O}_{\mathbb{P}^1}(-1).$$

Thus the Itaka dimension drops upon restriction. On the other hand, on $X = \mathbb{P}^1 \times \mathbb{P}^1$ take $L = \text{pr}_1^* \mathcal{O}_{\mathbb{P}^1}(-1) \otimes \text{pr}_2^* \mathcal{O}_{\mathbb{P}^1}(1)$. For $m > 0$,

$$H^0(X, L^{\otimes m}) \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-m)) \otimes H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(m)) = 0.$$

Hence $\kappa(X, L) = -\infty$, whereas for $Y = \{\text{point}\} \times \mathbb{P}^1$ we have $\kappa(Y, L|_Y) = 1$.

Example 3.1.7 (Non-normal varieties). Linear series can behave differently from their pullbacks to the normalization. Let X be a nodal plane cubic. By [Har77, II, Exercise 6.7], $\text{Pic}^0(X) \cong \mathbb{G}_m$. Choose a non-torsion line bundle L , corresponding to $\lambda \in \mathbb{C}^*$ which is not a root of unity. The normalization $\nu : \mathbb{P}^1 \rightarrow X$ identifies two points p, q , and $\nu^* L \cong \mathcal{O}_{\mathbb{P}^1}$. The descent data identify the fibres at p, q by multiplication by λ (after choosing trivializations). Thus the pullback of a section of $L^{\otimes m}$ is a constant c satisfying

$$s'(p) = \lambda^m s'(q), \quad \text{that is,} \quad c = \lambda^m c.$$

For every $m > 0$ this forces $c = 0$. Consequently

$$H^0(X, L^{\otimes m}) = 0 \quad (m > 0), \quad H^0(\mathbb{P}^1, (\nu^* L)^{\otimes m}) = \mathbb{C}.$$

Our normalization convention gives $\kappa(X, L) = 0$, although $N(X, L) = \{0\}$. This explains the normality hypothesis when Itaka dimension is described by the original spaces of sections.

Example 3.1.8. Assume that X is normal and $\kappa(X, L) = \kappa \geq 0$. Then $\dim \varphi_m(X) = \kappa$ for all sufficiently large $m \in N(X, L)$.

Proof. Choose $k > 0$ such that $\dim \varphi_k(X) = \kappa$. If $m \in N(L)$ is sufficiently large, then $m - k$ is a sufficiently large multiple of $e(L)$, and hence belongs to $N(L)$. Multiplication by a non-zero section of $L^{\otimes(m-k)}$ gives an injection

$$H^0(X, L^{\otimes k}) \hookrightarrow H^0(X, L^{\otimes m}).$$

The associated linear projection $\nu_{m-k} : \mathbb{P}H^0(X, L^{\otimes m}) \dashrightarrow \mathbb{P}H^0(X, L^{\otimes k})$ gives $\varphi_k = \nu_{m-k} \circ \varphi_m$. Therefore $\dim Y_m \geq \dim Y_k = \kappa$, and the reverse inequality is the definition. $\square$

Now we consider morphisms for which pullback preserves spaces of sections.

Lemma 3.1.9 (Coincidence for pullbacks). *Let $f : X \rightarrow Y$ be a projective surjective morphism of projective varieties satisfying $f_*\mathcal{O}_X = \mathcal{O}_Y$, and let L be a line bundle on Y . Then*

$$H^0(X, f^*(L^{\otimes m})) \cong H^0(Y, L^{\otimes m}) \quad (m \geq 0).$$

*In particular, $\kappa(Y, L) = \kappa(X, f^*L)$, with the normalization convention of Definition 3.1.3.*

Proof. The projection formula gives

$$f_*(f^*L^{\otimes m}) = f_*(\mathcal{O}_X \otimes f^*L^{\otimes m}) \cong f_*\mathcal{O}_X \otimes L^{\otimes m} \cong L^{\otimes m}.$$

Taking global sections proves the first assertion. In the normal case the complete linear-series maps have the same images, since f is surjective, which proves the assertion about Iitaka dimension in that case. For general X, Y , normalize both and take the Stein factorization of the induced morphism $X^\nu \rightarrow Y^\nu$. Its fibre-space part preserves Iitaka dimension by the normal case just proved, and its finite part preserves Iitaka dimension by finite-pullback invariance for normal projective varieties (see [Pop, Exercise 1.2.5]). This gives the stated identity with the normalization convention. $\square$

Definition 3.1.10 (Algebraic fibre space). An algebraic fibre space is a surjective projective morphism $f : X \rightarrow Y$ of varieties such that $f_*\mathcal{O}_X = \mathcal{O}_Y$.

For an arbitrary projective surjective morphism $g : V \rightarrow W$, Stein factorization [Har77, III, 11.5] expresses g as

$$V \xrightarrow{g'} W' \xrightarrow{h} W, \quad W' = \operatorname{Spec}_W(g_*\mathcal{O}_V),$$

where g' is an algebraic fibre space and h is finite. Thus g is a fibre space if and only if its finite Stein factor is trivial. The theorem on formal functions shows that the fibres of a fibre space are connected. Conversely, if Y is normal and $f : X \rightarrow Y$ has connected fibres, write $f = h \circ g$ for its Stein factorization. For a closed point $y \in Y$,

$$f^{-1}(y) = \coprod_{y' \in h^{-1}(y)} g'^{-1}(y').$$

Every set on the right is non-empty, so $h^{-1}(y)$ consists of one point. In characteristic zero a finite dominant map is generically separable, and on a dense open subset its fibres have $[\mathbb{C}(Y') : \mathbb{C}(Y)]$ points. Thus this degree is 1. The finite birational map h is an isomorphism because Y is normal. If $f : X \rightarrow Y$ is a fibre space, X is normal, and $\mu : X' \rightarrow X$ is a projective birational morphism, then $\mu_*\mathcal{O}_{X'} = \mathcal{O}_X$ and $(f \circ \mu)_*\mathcal{O}_{X'} = \mathcal{O}_Y$. Hence the composition is again a fibre space.

Example 3.1.11 (Fibre spaces and function fields). Let $f : X \rightarrow Y$ be a projective surjective morphism of normal projective varieties. Then f is an algebraic fibre space if and only if $\mathbb{C}(Y)$ is algebraically closed in $\mathbb{C}(X)$.

Proof. “ $\Rightarrow$ ”: Suppose not. There is a finite extension

$$\mathbb{C}(Y) \subsetneq K \subseteq \mathbb{C}(X), \quad [K : \mathbb{C}(Y)] \geq 2.$$

Choose a normal projective model Y' of K . The field inclusions give rational maps $X \dashrightarrow Y' \dashrightarrow Y$ factoring f , with the second map generically finite of degree at least 2. We eliminate indeterminacies in two steps, as in [Har77, II, 7.17.3].

Step 1: Embed Y in projective space and resolve the rational map $v : Y' \dashrightarrow Y$:

$$\begin{array}{ccc} \widetilde{Y}' & & \\ \pi_1 \downarrow & \searrow v' & \\ Y' & \dashrightarrow & Y \end{array}$$

Normalize the modification if necessary. We now have a rational map $X \dashrightarrow \widetilde{Y}'$.

Step 2: Embed $\widetilde{Y}'$ in projective space and resolve this map:

$$\begin{array}{ccc} \widetilde{X} & & \\ \pi_2 \downarrow & \searrow u' & \\ X & \dashrightarrow & \widetilde{Y}' \end{array}$$

Thus $f \circ \pi_2 = v' \circ u'$. Since X is normal, $\pi_{2*}\mathcal{O}_{\widetilde{X}} = \mathcal{O}_X$, so $\widetilde{X} \rightarrow Y$ remains an algebraic fibre space. On a dense open subset of Y , the map v' is finite étale of degree $[K : \mathbb{C}(Y)] \geq 2$. Its closed fibres have at least two points, and their inverse images under the proper surjective map u' disconnect the corresponding fibres of $\widetilde{X} \rightarrow Y$, a contradiction. Here it is the closed fibres on this open set (or the *geometric generic fibre*) that have several points; $\text{Spec } K$ itself has only one.

“ $\Leftarrow$ ”: Run the Stein factorization $X \rightarrow Y' \rightarrow Y$. The extension $\mathbb{C}(Y')/\mathbb{C}(Y)$ is finite and lies in $\mathbb{C}(X)$, hence is trivial. Since Y is normal, $Y' \rightarrow Y$ is an isomorphism. $\square$

The function-field criterion also makes sense in the birational category: a dominant rational map of normal varieties is a rational fibre space when the induced inclusion of function fields is algebraically closed.

Example 3.1.12 (Injectivity of Picard groups). Let X, Y be projective varieties and $f : X \rightarrow Y$ an algebraic fibre space. Then $f^* : \text{Pic}(Y) \rightarrow \text{Pic}(X)$ is injective.

Proof. Let $B = \mathcal{O}_Y(D)$ satisfy $f^*B \cong \mathcal{O}_X$. By Lemma 3.1.9,

$$H^0(Y, B) = H^0(X, f^*B) \neq 0, \quad H^0(Y, B^\vee) = H^0(X, f^*B^\vee) \neq 0.$$

There are effective Cartier divisors $E_1 \sim_{\text{lin}} D$ and $E_2 \sim_{\text{lin}} -D$. The effective divisor $E_1 + E_2$ is linearly equivalent to zero. Its defining section, viewed as a section of $\mathcal{O}_Y$, is a non-zero constant, so $E_1 + E_2 = 0$. Hence $E_1 = E_2 = 0$ and $D \sim_{\text{lin}} 0$. $\square$

Example 3.1.13 (Normality of fibre spaces). If $f : X \rightarrow Y$ is an algebraic fibre space and X is normal, then Y is normal.

Proof. The universal property of the normalization $\nu : Y' \rightarrow Y$ gives $f = \nu \circ g$ for a morphism $g : X \rightarrow Y'$. Since these maps are dominant,

$$\mathcal{O}_Y = (\nu \circ g)_* \mathcal{O}_X = \nu_* g_* \mathcal{O}_X \supseteq \nu_* \mathcal{O}_{Y'} \supseteq \mathcal{O}_Y.$$

Thus $\nu_* \mathcal{O}_{Y'} = \mathcal{O}_Y$, and, since ν is finite,

$$Y' = \operatorname{Spec}_Y(\nu_* \mathcal{O}_{Y'}) \cong \operatorname{Spec}_Y \mathcal{O}_Y = Y.$$

□

Example 3.1.14 (Exceptional twists of birational pullbacks). Let $\mu : X' \rightarrow X$ be a proper birational morphism of varieties, with X' normal. Let L be a line bundle on X , and E a μ -exceptional effective Cartier divisor on X' : every component of E maps to a subset of codimension at least 2 in X . Then the natural inclusion

$$H^0(X', \mu^* L) \longrightarrow H^0(X', \mu^* L \otimes \mathcal{O}_{X'}(E))$$

is an isomorphism.

Proof. Factor μ through the normalization $\nu : X^\nu \rightarrow X$, writing $\mu = \nu \circ \mu'$. For $L^\nu = \nu^* L$ we have $\mu^* L = (\mu')^* L^\nu$. Since ν is finite, the exceptional divisors for μ and μ' coincide. Thus we may assume X normal. Now $\mu_* \mathcal{O}_{X'} = \mathcal{O}_X$, and the projection formula gives

$$H^0(X', \mu^* L) \cong H^0(X, L).$$

Set $Z = \mu(\operatorname{Supp} E)$, $U = X \setminus Z$, and $U' = \mu^{-1}(U)$. A section of $\mu^* L(E)$ restricts to a section of $\mu^* L$ on U' , hence to a section of L on U . Choose open sets V_i trivializing L . By normality and $\operatorname{codim}_X Z \geq 2$, the restriction map

$$H^0(V_i, \mathcal{O}_{V_i}) \longrightarrow H^0(V_i \cap U, \mathcal{O}_{V_i})$$

is an isomorphism. These extensions glue uniquely to a section of L on X . Its pull-back agrees with the original section on the dense open U' and hence everywhere. This proves surjectivity of the natural inclusion. □

Definition 3.1.15 (Section ring associated to a line bundle). For a line bundle L on a projective variety X , its graded ring or section ring is the graded $\mathbb{C}$ -algebra

$$R(L) = R(X, L) = \bigoplus_{m \geq 0} H^0(X, L^{\otimes m}).$$

The section ring $R(D) = R(X, D)$ of a Cartier divisor is defined similarly.

Example 3.1.16. For $X = \mathbb{P}^n$ and $L = \mathcal{O}_{\mathbb{P}^n}(1)$, we have $R(L) = \mathbb{C}[T_0, \dots, T_n]$, the homogeneous coordinate ring of $\mathbb{P}^n$.

The section ring is finitely generated in this example, but need not be so in general. This property greatly influences the geometry of L .

Definition 3.1.17 (Finitely generated line bundles and Cartier divisors). A line bundle L on a projective variety is finitely generated if $R(X, L)$ is a finitely generated $\mathbb{C}$ -algebra. A Cartier divisor D is finitely generated if $\mathcal{O}_X(D)$ is so.

Passing to additive notation, let D be a Cartier divisor on a projective variety X . We consider the set-theoretic base locus

$$\text{Bs}(|mD|) = \bigcap_{D' \in |mD|} \text{Supp}(D'),$$

with $\text{Bs}(|mD|) = X$ if $|mD| = \emptyset$. As in Chapter 1, this is the base scheme equipped with its reduced structure.

Definition 3.1.18 (Stable base locus). The stable base locus of D is the algebraic set

$$\mathbf{B}(D) = \bigcap_{m \geq 1} \text{Bs}(|mD|).$$

The intersection is set-theoretic, rather than an intersection of schemes.

Proposition 3.1.19. *The stable base locus is the unique minimal element of $\{\text{Bs}(|mD|)\}_{m \geq 1}$. In particular, there exists $m_0 > 0$ such that*

$$\mathbf{B}(D) = \text{Bs}(|km_0D|) \quad \text{for all } k \geq 1.$$

Proof. For $m, l \geq 1$, multiplication of sections gives $\mathfrak{b}(|mD|)^l \subseteq \mathfrak{b}(|lmD|)$, hence $\text{Bs}(|lmD|) \subseteq \text{Bs}(|mD|)$. Since X is noetherian, the family of closed sets has a minimal element, say $\text{Bs}(|m_0D|)$. For every m ,

$$\text{Bs}(|mm_0D|) \subseteq \text{Bs}(|m_0D|) \cap \text{Bs}(|mD|).$$

Minimality makes the first inclusion into $\text{Bs}(|m_0D|)$ an equality. Thus $\text{Bs}(|m_0D|)$ lies in every member of the family, and equals $\mathbf{B}(D)$. The same argument gives $\text{Bs}(|km_0D|) = \mathbf{B}(D)$ for every $k \geq 1$. $\square$

Remark 3.1.20 (Realizing the stable base locus). It is not always possible to take $m_0 = 1$. If $N(D) \neq \{0\}$ and $e(D) = e > 1$, then $\mathbf{B}(D) \neq X$, whereas $\text{Bs}(|mD|) = X$ whenever $e \nmid m$. Thus equality $\mathbf{B}(D) = \text{Bs}(|mD|)$ is possible only when $e \mid m$. Even when $e(D) = 1$, equality can fail for arbitrarily large m .

Example 3.1.21 (Multiples). For every Cartier divisor D and every integer $p \geq 1$, $\mathbf{B}(pD) = \mathbf{B}(D)$.

Proof. By Proposition 3.1.19 choose $m_0, m'_0 > 0$ such that

$$\mathbf{B}(D) = \text{Bs}(|km_0D|), \quad \mathbf{B}(pD) = \text{Bs}(|k'm'_0pD|)$$

for positive k, k' . Taking $k = m'_0p$ and $k' = m_0$ identifies the two sets. $\square$

Remark 3.1.22 ($\mathbb{Q}$ -Cartier divisors). For a $\mathbb{Q}$ -Cartier divisor D , choose $p > 0$ such that pD is an integral Cartier divisor and define $\mathbf{B}(D) = \mathbf{B}(pD)$. The preceding example shows that this is independent of p .

Example 3.1.23 (Scheme structures on the base locus). Although the underlying sets can stabilize, the base schemes need not. Let $X = \text{Bl}_p \mathbb{P}^2$, with exceptional divisor E and pullback of a line H , and set $D = H + E$. We have $|mD| = |mH| + mE$. Indeed, the exact sequence

$$0 \longrightarrow \mathcal{O}_X(mD - E) \xrightarrow{\cdot s_E} \mathcal{O}_X(mD) \longrightarrow \mathcal{O}_E(mD|_E) \longrightarrow 0$$

and $(mD \cdot E) = -m < 0$ show that multiplication by s_E induces an isomorphism on global sections. Consequently $|mD| = |mD - E| + E$. Repeating the argument until all m copies of E have been removed gives $|mD| = |mH| + mE$, with $|mH|$ free. The evaluation map defining the base ideal therefore gives

$$\mathfrak{b}(|mD|) = \mathcal{O}_X(-mE) \mathfrak{b}(|mH|) = \mathcal{O}_X(-mE).$$

The base scheme is the thickening mE defined by $\mathcal{I}_E^m$, so these schemes are distinct even though their underlying set is always E .

3.1B Semiample Line Bundles

We are investigating the asymptotic behavior of the mappings φ_k determined by $|L^{\otimes k}|$. This subsection focuses on the important case where some power of L is free. On a complete scheme the spaces of global sections are finite-dimensional, so the corresponding projective spaces are defined.

Definition 3.1.24. A line bundle L on a complete scheme is *semi-ample* if $L^{\otimes m}$ is globally generated, that is, $|L^{\otimes m}|$ is free, for some $m > 0$. A Cartier divisor D is semi-ample if $\mathcal{O}_X(D)$ is so.

Fix a semi-ample line bundle L on a projective variety, and write

$$M(X, L) = \{m \in \mathbb{Z}_{\geq 0} \mid L^{\otimes m} \text{ is free}\} \subseteq N(X, L).$$

Let $f = f(L)$ be the greatest common divisor of its positive elements. By the semigroup observation following Definition 3.1.1, $L^{\otimes kf}$ is free for all $k \gg 0$. For $m > 0$ in $M(X, L)$, write

$$\varphi_m : X \longrightarrow Y_m \subseteq \mathbb{P}H^0(X, L^{\otimes m})$$

for the map of the complete linear series, viewed as a morphism onto its image.

Theorem 3.1.25 (Semi-ample fibrations). *Let X be a normal projective variety and L a semi-ample line bundle. There is an algebraic fibre space $\varphi : X \rightarrow Y$ such that, for every sufficiently large $m \in M(X, L)$, $Y_m \cong Y$ and $\varphi_m = \varphi$ up to this isomorphism. Furthermore, there is an ample line bundle A on Y such that $\varphi^* A \cong L^{\otimes f}$, where $f = f(L)$.*

Thus the morphisms stabilize to a fibre space structure. The bundle $L^{\otimes f}$ is trivial on each fibre, since

$$L^{\otimes f}|_{X_y} \cong (\varphi|_{X_y})^*(A \otimes k(y)) \cong \mathcal{O}_{X_y}.$$

By Lemma 3.1.9 and Example 3.1.13, Y is normal and

$$H^0(X, L^{\otimes kf}) \cong H^0(Y, A^{\otimes k}) \quad (k \geq 0).$$

Before proving the theorem, consider the multiplication map

$$\mu_k : \text{Sym}^k H^0(X, L^{\otimes m}) \longrightarrow H^0(X, L^{\otimes km}).$$

Its image W_k determines a free subseries giving the k th Veronese re-embedding of Y_m . The linear projection associated to $W_k \subseteq H^0(X, L^{\otimes km})$, followed by the inverse of that Veronese embedding on its image, gives a finite morphism $\nu_k : Y_{km} \rightarrow Y_m$ such that

$$\begin{array}{ccc} X & \xrightarrow{\varphi_{km}} & Y_{km} \\ \varphi_m \downarrow & \swarrow \nu_k & \\ Y_m & & \end{array}$$

Finiteness follows because the projection is defined everywhere on Y_{km} and pulls back the ample hyperplane bundle on its image to the ample hyperplane bundle of Y_{km} , so it contracts no positive-dimensional fibre. The very ample bundle $A_m = \mathcal{O}_{\mathbb{P}}(1)|_{Y_m}$ satisfies $\varphi_m^* A_m \cong L^{\otimes m}$. Also

$$H^0(Y_m, A_m) \cong H^0(X, L^{\otimes m}).$$

Indeed, restriction from the ambient projective space and pullback to X are injective, and their composition is the identity on the space of sections defining this *complete* linear series. This equality does not yet require φ_m to be a fibre space.

Lemma 3.1.26. *In the situation of Theorem 3.1.25, fix $m > 0$ in $M(X, L)$. For all sufficiently large integers k , the factorization*

$$X \xrightarrow{\varphi_{km}} Y_{km} \xrightarrow{\nu_k} Y_m$$

is the Stein factorization of φ_m . In particular, Y_{km} and φ_{km} are independent of k up to isomorphism for $k \gg 0$.

Proof. Let $X \xrightarrow{\psi} V \xrightarrow{\mu} Y_m$ be the Stein factorization. The variety V is normal, and μ is finite. Hence $B = \mu^* A_m$ is ample, and $B^{\otimes k}$ is very ample for $k \gg 0$. Since

$$\psi^* B^{\otimes k} \cong L^{\otimes km}, \quad H^0(X, L^{\otimes km}) \cong H^0(V, B^{\otimes k}),$$

we can choose a basis $t_0, \dots, t_N$ of the latter space and write

$$\begin{aligned} \varphi_{km}(x) &= [\psi^* t_0(x) : \dots : \psi^* t_N(x)] \\ &= [t_0(\psi(x)) : \dots : t_N(\psi(x))] = \varphi_{|B^{\otimes k}|}(\psi(x)). \end{aligned}$$

The last map embeds V in projective space. Thus $Y_{km} \cong V$ and φ_{km} is ψ up to this isomorphism. $\square$

Proof of Theorem 3.1.25. Replace L by $L^{\otimes f}$ to assume $f = 1$, so every sufficiently large power of L is free. Choose relatively prime large integers a, b . By Lemma 3.1.26 there are integers K_a, K_b such that the maps φ_{ka} , respectively φ_{kb} , stabilize for $k \geq K_a$, respectively $k \geq K_b$. Choose distinct primes $l_a \geq K_a$, $l_b \geq K_b$ not dividing ab , and set $p = l_a a$, $q = l_b b$. Then $\gcd(p, q) = 1$, and

$$\varphi_{kp} = \varphi_p, \quad \varphi_{kq} = \varphi_q \quad (k \geq 1)$$

up to isomorphism. In particular, comparison at the common multiple pq gives $Y_p \cong Y_{pq} \cong Y_q$. Denote the resulting fibre space by $\varphi : X \rightarrow Y$. The variety Y carries very ample bundles A_p, A_q with $\varphi^* A_p = L^{\otimes p}$ and $\varphi^* A_q = L^{\otimes q}$. Choose integers r, s with $rp + sq = 1$, and set

$$A = A_p^{\otimes r} \otimes A_q^{\otimes s}.$$

Then $\varphi^* A = L$. The injectivity of $\varphi^* : \text{Pic}(Y) \rightarrow \text{Pic}(X)$ gives $A^{\otimes p} \cong A_p$ and $A^{\otimes q} \cong A_q$. In particular, A is ample. It remains to identify all sufficiently large exponents. For $c, d \geq 1$, the product map

$$\text{Sym}^c H^0(Y, A^{\otimes p}) \otimes \text{Sym}^d H^0(Y, A^{\otimes q}) \longrightarrow H^0(Y, A^{\otimes(cp+dq)})$$

gives a subseries embedding Y by the corresponding Segre–Veronese map. Moreover, Lemma 3.1.9 identifies this complete space of sections with $H^0(X, L^{\otimes(cp+dq)})$. Thus φ_{cp+dq} is φ followed by the embedding associated to the complete series on Y . Every sufficiently large integer is of the form $cp + dq$ with $c, d \geq 1$. Equivalently, one can use directly that $A^{\otimes m}$ is very ample for $m \gg 0$. Restoring the original L gives $\varphi^* A = L^{\otimes f}$ and the required conclusion for all sufficiently large $m \in M(X, L)$. $\square$

Semi-ample fibrations lead to finite generation properties.

Example 3.1.27 (Surjectivity of multiplication maps). Let X be a normal projective variety and L a globally generated line bundle. There exists $m_0 = m_0(L)$ such that

$$H^0(X, L^{\otimes a}) \otimes H^0(X, L^{\otimes b}) \longrightarrow H^0(X, L^{\otimes(a+b)})$$

is surjective for $a, b \geq m_0$. More generally, for any coherent sheaf $\mathcal{F}$,

$$H^0(X, \mathcal{F} \otimes L^{\otimes a}) \otimes H^0(X, L^{\otimes b}) \longrightarrow H^0(X, \mathcal{F} \otimes L^{\otimes(a+b)})$$

is surjective for $a, b \gg 0$.

Proof. Take the Stein factorization of the map associated to $|L|$:

$$X \xrightarrow{\varphi} Y \xrightarrow{\nu} Y_1 \subseteq \mathbb{P}H^0(X, L).$$

The bundle $A = \nu^*\mathcal{O}_{Y_1}(1)$ is both ample and globally generated, and $\varphi^*A = L$. The projection formula gives

$$\varphi_*(\mathcal{F} \otimes L^{\otimes m}) \cong \varphi_*\mathcal{F} \otimes A^{\otimes m}.$$

Thus it suffices to prove surjectivity of

$$H^0(Y, \mathcal{G} \otimes A^{\otimes a}) \otimes H^0(Y, A^{\otimes b}) \longrightarrow H^0(Y, \mathcal{G} \otimes A^{\otimes(a+b)}), \quad \mathcal{G} = \varphi_*\mathcal{F}.$$

The sheaf $\mathcal{G}$ is coherent. For some m_0 it is m_0 -regular with respect to the ample globally generated bundle A :

$$H^i(Y, \mathcal{G} \otimes A^{\otimes(m_0-i)}) = 0 \quad (i > 0).$$

Indeed, for $i > \dim Y$ this is Grothendieck vanishing; for the remaining finitely many i , choose $m_0 \geq n_0 + \dim Y$ using Serre vanishing. Castelnuovo–Mumford regularity [Laz04a, Theorem 1.8.5] now gives the surjectivity of multiplication by $H^0(Y, A)$ for $a \geq m_0$. Iterating, the product with $H^0(Y, A)^{\otimes b}$ is onto for every $b \geq 0$. It factors through the displayed multiplication map with $H^0(Y, A^{\otimes b})$, which is therefore also onto. Taking $\mathcal{F} = \mathcal{O}_X$ gives the first assertion. $\square$

Theorem 3.1.28 (Finite generation of semi-ample line bundles). *Let L be a semi-ample line bundle on a normal projective variety X . Then $R(X, L)$ is a finitely generated $\mathbb{C}$ -algebra.*

Proof. Choose $k > 0$ such that $L^{\otimes k}$ is free. By Example 3.1.27,

$$H^0(X, L^{\otimes ka}) \otimes H^0(X, L^{\otimes kb}) \longrightarrow H^0(X, L^{\otimes k(a+b)})$$

is onto for $a, b \geq m_0$. Thus the Veronese subalgebra

$$R(X, L)^{(k)} = \bigoplus_{m \geq 0} H^0(X, L^{\otimes km})$$

is generated by its pieces of degrees at most $2m_0 - 1$: every higher piece is a product of the piece of degree m_0 with one of lower degree. For $0 \leq i < k$, apply

the sheaf version of Example 3.1.27 to the *free* bundle $L^{\otimes k}$ and the coherent sheaf $L^{\otimes i}$. It shows that

$$\bigoplus_{m \geq 0} H^0(X, L^{\otimes i} \otimes (L^{\otimes k})^{\otimes m})$$

is a finitely generated module over $R(X, L)^{(k)}$ (fix a sufficiently large degree in the multiplication map and induct on m). Their finite direct sum is $R(X, L)$, so this ring is finitely generated. $\square$

Zariski proved a striking result about finite base loci in [Zar62, II, 6.2], and Fujita generalized it in [Fuj80].

Theorem 3.1.29 (Zariski–Fujita). *Let L be a line bundle on a projective variety X such that $\text{Bs}(|L|)$ is a finite set. Then L is semi-ample: $L^{\otimes m}$ is free for some $m > 0$.*

3.1C Iitaka Fibration

We now give the asymptotic analysis of the rational mappings

$$\varphi_k = \varphi_{|L^{\otimes k}|} : X \dashrightarrow Y_k \subseteq \mathbb{P}H^0(X, L^{\otimes k})$$

in the general setting, always viewing φ_k as a rational map to Y_k .

Theorem 3.1.30 (Iitaka fibrations). *Let X be a normal projective variety, and L a line bundle such that $\kappa(X, L) > 0$. For all sufficiently large $k \in N(X, L)$, the rational maps $\varphi_k : X \dashrightarrow Y_k$ are birationally equivalent to a fixed algebraic fibre space $\varphi_\infty : X_\infty \rightarrow Y_\infty$ of normal projective varieties. More precisely, there is a commutative diagram*

$$\begin{array}{ccc} X & \xleftarrow{u_\infty} & X_\infty \\ \varphi_k \downarrow & & \downarrow \varphi_\infty \\ Y_k & \xleftarrow{\nu_k} & Y_\infty \end{array}$$

where u_∞ is a birational morphism and ν_k is birational. One has $\dim Y_\infty = \kappa(X, L)$. Setting $L_\infty = u_\infty^* L$, a very general fibre F of φ_∞ satisfies $\kappa(F, L_\infty|_F) = 0$.

Definition 3.1.31 (Iitaka fibration associated to a line bundle). The morphism $\varphi_\infty : X_\infty \rightarrow Y_\infty$ in the theorem is an Iitaka fibration associated to L . It is unique up to birational equivalence, whereas a semi-ample fibration is unique up to isomorphism. For a Cartier divisor D use the line bundle $\mathcal{O}_X(D)$. When $\kappa(X, L) = 0$, the Iitaka fibration is the map to a point.

Definition 3.1.32 (Iitaka fibration of a variety). For a variety X of non-negative Kodaira dimension, its Iitaka fibration is the Iitaka fibration associated to the canonical bundle on a smooth projective model. When $\kappa(X) = -\infty$, this construction does not produce an Iitaka fibration.

Proof of Theorem 3.1.30. Fix $m > 0$ in $N(X, L)$ with $\dim Y_m = \kappa(X, L)$.

Step 1: We first prove that, for $k \gg 0$, all maps φ_{km} are birationally equivalent to a fixed algebraic fibre space $\psi_{(m)} : X_{(m)} \rightarrow Y_{(m)}$ of normal varieties. Blow up the base ideal $\mathfrak{b} = \mathfrak{b}(|L^{\otimes m}|)$:

$$\varepsilon : \widetilde{X} = \text{Proj}_X \left(\bigoplus_{d \geq 0} \mathfrak{b}^d \right) \rightarrow X.$$

The inverse-image ideal is invertible, $\varepsilon^{-1}\mathfrak{b} \cdot \mathcal{O}_{\widetilde{X}} = \mathcal{O}_{\widetilde{X}}(-E)$, where E is an effective Cartier divisor. It need not be ε -exceptional: the original series can already have divisorial fixed components. Normalize, writing $\nu : X_{(m)} \rightarrow \widetilde{X}$ and $u_m = \varepsilon \circ \nu$. The evaluation map gives a surjection

$$H^0(X, L^{\otimes m}) \otimes \mathcal{O}_X \rightarrow \mathfrak{b} \otimes L^{\otimes m}.$$

Pulling back and then taking the image of $u_m^* \mathfrak{b} \rightarrow \mathcal{O}_{X_{(m)}}$ gives a surjection onto $(u_m^{-1} \mathfrak{b} \cdot \mathcal{O}_{X_{(m)}}) \otimes u_m^* L^{\otimes m}$. Here

$$\begin{aligned} u_m^{-1} \mathfrak{b} \cdot \mathcal{O}_{X_{(m)}} &= \nu^{-1}(\varepsilon^{-1} \mathfrak{b} \cdot \mathcal{O}_{\widetilde{X}}) \cdot \mathcal{O}_{X_{(m)}} \\ &= \nu^* \mathcal{O}_{\widetilde{X}}(-E) = \mathcal{O}_{X_{(m)}}(-F_m), \quad F_m = \nu^* E. \end{aligned}$$

The pullback of a local equation of E remains non-zero on the integral variety $X_{(m)}$, because ν is dominant. Thus it defines the indicated effective Cartier divisor; no flatness of ν is being assumed. Set

$$M_m = \mathcal{O}_{X_{(m)}}(-F_m) \otimes u_m^* L^{\otimes m}.$$

This line bundle is globally generated. Since X is normal, $H^0(X_{(m)}, u_m^* L^{\otimes m}) = H^0(X, L^{\otimes m})$, and all these sections vanish along F_m . Consequently

$$u_m^* |L^{\otimes m}| = |M_m| + F_m, \quad H^0(X_{(m)}, M_m) = H^0(X, L^{\otimes m}).$$

The morphism $\psi_m : X_{(m)} \rightarrow Y_m$ defined by $|M_m|$ resolves φ_m . For the morphism ψ_{km} defined by $|M_m^{\otimes k}|$, we obtain the diagram

$$\begin{array}{ccc} X & \xleftarrow{u_m} & X_{(m)} \\ \varphi_m \downarrow \text{dashed} & \searrow \psi_m & \downarrow \psi_{km} \\ Y_m & \xleftarrow{\lambda_k} & Y'_{km} \end{array}$$

where λ_k is finite. By semi-ample fibrations, ψ_{km} stabilizes for $k \gg 0$ to a fibre space $\psi_{(m)} : X_{(m)} \rightarrow Y_{(m)}$. On the other hand, multiplication by the section of kF_m embeds $H^0(X_{(m)}, M_m^{\otimes k})$ in $H^0(X, L^{\otimes km})$. Thus, as rational maps,

$$\psi_{(m)} = \mu_k \circ \varphi_{km} \circ u_m, \quad \mu_k : Y_{km} \dashrightarrow Y_{(m)}.$$

Since λ_k is finite, $\dim Y_{(m)} = \dim Y_m = \kappa(X, L)$, and hence

$$\dim Y_{(m)} \leq \dim Y_{km} \leq \kappa(X, L) = \dim Y_{(m)}.$$

Therefore μ_k is generically finite. The inclusion

$$\mathbb{C}(Y_{(m)}) \subseteq \mathbb{C}(Y_{km}) \subseteq \mathbb{C}(X)$$

and the algebraic closedness of $\mathbb{C}(Y_{(m)})$ in $\mathbb{C}(X)$ force its degree to be 1. Thus μ_k is birational, proving Step 1.

Step 2: We show that these fibre spaces are birationally isomorphic and choose a common model. Replacing L by $L^{\otimes e}$, with $e = e(L)$, we may assume $e = 1$. Positive powers do not change Iitaka dimension: products of sections give Veronese subseries with the same image dimension. We restore the original line bundle at the end of the proof. Choose relatively prime integers $p, q > 1$, both sufficiently large that $\dim Y_p = \dim Y_q = \kappa(X, L)$. Choose $m \gg 0$ so that the powers $M_p^{\otimes p^{m-1}}$ and $M_q^{\otimes q^{m-1}}$ give the stabilized fibre spaces $\psi_{(p)}$ and $\psi_{(q)}$ of Step 1.

To construct a common resolution, form $Z = X_{(p)} \times_X X_{(q)}$. On a dense open $U \subseteq X$ over which u_p, u_q are isomorphisms, $Z_U \cong U$. Take the reduced closure of Z_U in Z and normalize it. This gives a normal projective variety X_∞ and birational morphisms v_p, v_q with $u_p \circ v_p = u_q \circ v_q =: u_\infty$:

$$\begin{array}{ccc} X_\infty & \xrightarrow{v_p} & X_{(p)} \\ v_q \downarrow & & \downarrow u_p \\ X_{(q)} & \xrightarrow{u_q} & X \end{array}$$

Consider the globally generated line bundle

$$M_{p,q} = v_p^* M_p^{\otimes p^{m-1}} \otimes v_q^* M_q^{\otimes q^{m-1}}.$$

By construction, these two factors are pulled back from very ample bundles $\mathcal{O}_{Y_{(p)}}(1)$ and $\mathcal{O}_{Y_{(q)}}(1)$ for the chosen stabilized embeddings. Their external tensor product is very ample on $Y_{(p)} \times Y_{(q)}$, and its pullback by

$$\pi = (\psi_{(p)} \circ v_p, \psi_{(q)} \circ v_q) : X_\infty \longrightarrow Y_{(p)} \times Y_{(q)}$$

is $M_{p,q}$. Let Y_∞ be the normalization of the image of the *complete* series $|M_{p,q}|$, and let $\varphi_\infty : X_\infty \rightarrow Y_\infty$ be the induced morphism. The map from the complete-series image to $\pi(X_\infty)$ is finite: it pulls back the ample hyperplane bundle to

the hyperplane bundle. Thus Y_∞ maps finitely onto $\pi(X_\infty)$, which is a closed subvariety of the product. We obtain

$$\begin{array}{ccccc}
 X_{(p)} & \xleftarrow{v_p} & X_\infty & \xrightarrow{v_q} & X_{(q)} \\
 \psi_{(p)} \downarrow & & \downarrow \varphi_\infty & & \downarrow \psi_{(q)} \\
 Y_{(p)} & \xleftarrow{w_p} & Y_\infty & \xrightarrow{w_q} & Y_{(q)}
 \end{array}$$

Both w_p, w_q are surjective, so $\dim Y_\infty \geq \kappa(X, L)$. On the other hand,

$$M_{p,q} \hookrightarrow u_\infty^* L^{\otimes(p^m+q^m)},$$

and its complete series is a subseries of the latter complete series. Hence $\dim Y_\infty \leq \dim Y_{p^m+q^m} \leq \kappa(X, L)$. Thus equality holds and w_p, w_q are generically finite. The maps $\psi_{(p)} \circ v_p$ and $\psi_{(q)} \circ v_q$ are fibre spaces, because v_p, v_q are birational morphisms to normal varieties. The function-field criterion now makes w_p, w_q birational. It also shows that φ_∞ is a fibre space: $\mathbb{C}(Y_\infty) = \mathbb{C}(Y_{(p)})$ is algebraically closed in $\mathbb{C}(X_\infty)$. By construction, Y_∞ carries an ample globally generated bundle $A_{p,q}$ with $\varphi_\infty^* A_{p,q} = M_{p,q}$.

For $c, d \geq 1$, multiply by a fixed non-zero section of $v_p^* M_p^{\otimes(c-1)p^{m-1}} \otimes v_q^* M_q^{\otimes(d-1)q^{m-1}}$ to obtain

$$\begin{aligned}
 H^0(X_\infty, M_{p,q}) &\hookrightarrow H^0(X_\infty, v_p^* M_p^{\otimes cp^{m-1}} \otimes v_q^* M_q^{\otimes dq^{m-1}}) \\
 &\hookrightarrow H^0(X_\infty, u_\infty^* L^{\otimes(cp^m+dq^m)}).
 \end{aligned}$$

It follows that φ_∞ factors rationally through $\varphi_{cp^m+dq^m} \circ u_\infty$, with a generically finite map $\mu_{cp^m+dq^m} : Y_{cp^m+dq^m} \dashrightarrow Y_\infty$. As before, this map is birational because φ_∞ is a fibre space. Since $\gcd(p^m, q^m) = 1$, every sufficiently large integer has the form $cp^m + dq^m$ with $c, d \geq 1$. Taking $\nu_k = \mu_k^{-1}$ gives the diagram in the theorem.

Step 3: It remains to show that a very general fibre F has $\kappa(F, L_\infty|_F) = 0$, where $\overline{L_\infty} = u_\infty^* L$. Choose a very general closed point $y \in Y_\infty$ and set $F = \varphi_\infty^{-1}(y)$. We impose the following countably many open conditions. The fibre F is normal and integral; each ν_j with $j \gg 0$ is defined at y ; F is not contained in the base locus of any non-empty $|L_\infty^{\otimes j}|$; and cohomology and base change holds at y for every sheaf $L_\infty^{\otimes k}$, $k \geq 1$. The first condition holds on a dense open set: the generic fibre of a normal variety is normal, and a characteristic-zero fibre space has geometrically integral generic fibre. Generic flatness and cohomology and base change give the other required open sets. Over $\mathbb{C}$, their countable intersection contains very general points.

The map $\varphi_j \circ u_\infty$ is constant on the domain where it is defined in F . Thus all restricted sections are multiples of a single non-zero section s_F :

$$s_i|_F = c_i s_F.$$

This section may have zeros. Consequently

$$\rho_j : H^0(X_\infty, L_\infty^{\otimes j}) \longrightarrow H^0(F, L_\infty^{\otimes j}|_F)$$

has rank one for all sufficiently large j (we are still in the case $e(L) = 1$). In particular, some positive power of $L_\infty|_F$ has a non-zero section, giving $\kappa(F, L_\infty|_F) \geq 0$.

Choose a very ample line bundle B on Y_∞ . We claim that there is $m_0 > 0$ and a non-zero section

$$\sigma \in H^0(X_\infty, L_\infty^{\otimes m_0} \otimes \varphi_\infty^* B^\vee).$$

Choose F also so that $\sigma|_F \neq 0$; this is one more dense open condition. For fixed $k \geq 1$ and $r > 0$, multiplication by σ^r and restriction give the commutative square

$$\begin{array}{ccc} H^0(X_\infty, L_\infty^k \otimes \varphi_\infty^* B^r) & \xrightarrow{\cdot \sigma^r} & H^0(X_\infty, L_\infty^{k+rm_0}) \\ \beta_{k,r} \downarrow & & \downarrow \rho_{k+rm_0} \\ H^0(F, (L_\infty^k \otimes \varphi_\infty^* B^r)|_F) & \xrightarrow{\cdot \sigma^r|_F} & H^0(F, L_\infty^{k+rm_0}|_F) \end{array}$$

The lower horizontal map is injective because F is integral and $\sigma|_F$ is non-zero. Let $\mathcal{G}_k = \varphi_{\infty*}(L_\infty^{\otimes k})$. The projection formula and the base-change condition identify $\beta_{k,r}$ with evaluation

$$\begin{aligned} H^0(X_\infty, L_\infty^{\otimes k} \otimes \varphi_\infty^* B^{\otimes r}) &= H^0(Y_\infty, \mathcal{G}_k \otimes B^{\otimes r}) \\ &\longrightarrow (\mathcal{G}_k \otimes B^{\otimes r}) \otimes k(y) \\ &\cong H^0(F, L_\infty^{\otimes k}|_F) \otimes B^{\otimes r}|_y. \end{aligned}$$

For fixed k and $r \gg 0$, $\mathcal{G}_k \otimes B^{\otimes r}$ is globally generated. Its evaluation map is

$$H^0(Y_\infty, \mathcal{G}_k \otimes B^{\otimes r}) \otimes \mathcal{O}_{Y_\infty} \twoheadrightarrow \mathcal{G}_k \otimes B^{\otimes r}.$$

Tensoring its stalk at y with $k(y) = \mathbb{C}$ proves that $\beta_{k,r}$ is surjective. The right vertical map has rank one for $r \gg 0$, so the square and the lower injection imply $\text{rank } \beta_{k,r} \leq 1$. Hence

$$h^0(F, L_\infty^{\otimes k}|_F) \leq 1 \quad \text{for every } k \geq 1.$$

Together with the non-vanishing of some positive power this gives the desired Iitaka dimension zero.

It remains to prove the claim. The ample bundle $A = A_{p,q}$ satisfies $\varphi_\infty^* A = M_{p,q}$. For some $m_1 \gg 0$, $H^0(Y_\infty, A^{\otimes m_1} \otimes B^\vee) \neq 0$. Pull back a non-zero section and use

$$M_{p,q}^{\otimes m_1} \hookrightarrow L_\infty^{\otimes (p^m + q^m)m_1}$$

to obtain σ , with $m_0 = (p^m + q^m)m_1$. Finally restore the original L : the maps for $L^{\otimes e}$ are exactly the maps for the original exponents divisible by e , which include all sufficiently large elements of $N(L)$. On the fibre, taking a positive power does not change Iitaka dimension. This completes the proof. $\square$

Remark 3.1.33. At the beginning of the proof we removed the base locus of a linear system, a useful technique; see [Kaw24, 1.2.5]. We give the general version below.

Example 3.1.34. Let $|V|$ be a non-empty linear series of Cartier divisors on a normal variety X . There is a projective birational morphism $f : Y \rightarrow X$ from a normal variety such that

$$f^*|V| = |V_1| + F,$$

where F is an effective fixed Cartier divisor and $|V_1|$ is free.

Proof. Let $0 \neq V \subseteq H^0(X, \mathcal{O}_X(D))$ be the finite-dimensional space of sections. Write the image of evaluation as

$$\mathrm{im}(V \otimes \mathcal{O}_X \rightarrow \mathcal{O}_X(D)) = \mathfrak{b} \otimes \mathcal{O}_X(D).$$

Let $\varepsilon : \widetilde{X} = \mathrm{Bl}_{\mathfrak{b}} X \rightarrow X$ be the blow-up, $\nu : Y \rightarrow \widetilde{X}$ its normalization, and $f = \varepsilon \circ \nu$. Then

$$f^{-1}\mathfrak{b} \cdot \mathcal{O}_Y = \nu^{-1}(\varepsilon^{-1}\mathfrak{b} \cdot \mathcal{O}_{\widetilde{X}}) \cdot \mathcal{O}_Y = \nu^*\mathcal{O}_{\widetilde{X}}(-E) = \mathcal{O}_Y(-F).$$

Here F is effective Cartier, as in Step 1 above. To see projectivity explicitly, if $\dim V = n + 1$, the evaluation surjection induces a surjection of graded algebras

$$\mathrm{Sym}(\mathcal{O}_X^{\oplus(n+1)}) \twoheadrightarrow \bigoplus_{d \geq 0} \mathfrak{b}^d \otimes \mathcal{O}_X(dD).$$

Twisting each degree by $\mathcal{O}_X(dD)$ does not change relative Proj, so

$$\mathrm{Bl}_{\mathfrak{b}} X = \mathrm{Proj}_X \left(\bigoplus_{d \geq 0} \mathfrak{b}^d \right) \cong \mathrm{Proj}_X \left(\bigoplus_{d \geq 0} \mathfrak{b}^d \otimes \mathcal{O}_X(dD) \right) \hookrightarrow \mathbb{P}_X^n.$$

Thus ε , and hence f , is projective. The pullback evaluation map now gives

$$V \otimes \mathcal{O}_Y \twoheadrightarrow \mathcal{O}_Y(f^*D - F),$$

so the corresponding residual series $|V_1|$ is free and F is the fixed part of $f^*|V|$. $\square$

Corollary 3.1.35. *Let L be a line bundle on a normal projective variety X , and suppose $\kappa = \kappa(X, L) \geq 0$. There are constants $a, A > 0$ such that*

$$am^\kappa \leq h^0(X, L^{\otimes m}) \leq Am^\kappa$$

for all sufficiently large $m \in N(X, L)$.

Proof. If $\kappa = 0$, every non-empty positive complete linear series has projective dimension zero, so $h^0(X, L^{\otimes m}) = 1$ for positive $m \in N(L)$, and the assertion follows. Assume $\kappa > 0$. Spaces of sections are unchanged by a birational modification of a normal variety. We may therefore use the Iitaka fibration $\varphi : X \rightarrow Y$ and, by

resolving X , also assume X nonsingular. Step 3 of the theorem gives an ample line bundle B on Y and $m_0 > 0$ with $H^0(X, L^{\otimes m_0} \otimes \varphi^* B^\vee) \neq 0$. Multiplying by powers of this section gives

$$H^0(Y, B^{\otimes l}) = H^0(X, \varphi^* B^{\otimes l}) \hookrightarrow H^0(X, L^{\otimes l m_0}).$$

Asymptotic Riemann–Roch and Serre vanishing give

$$h^0(Y, B^{\otimes l}) = \frac{l^\kappa}{\kappa!} \int_Y c_1(B)^\kappa + O(l^{\kappa-1}).$$

The leading coefficient is positive, proving the lower bound along multiples of m_0 . To cover all sufficiently large $m \in N(L)$, let $e = e(L)$; necessarily $e \mid m_0$. Choose a sufficiently large multiple n of e such that $n+r \in N(L)$ for $r \in \{0, e, \dots, m_0 - e\}$. Divide

$$m - n = l m_0 + r, \quad 0 \leq r < m_0.$$

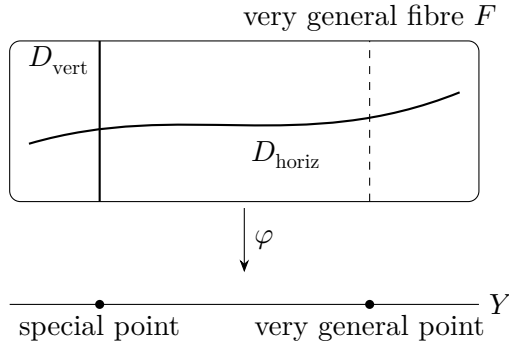
Then $e \mid r$, and multiplying by a non-zero section of $L^{\otimes(n+r)}$ gives $h^0(X, L^{\otimes m}) \geq h^0(X, L^{\otimes l m_0}) \geq C l^\kappa$. Since $l = (m - n - r)/m_0$ and there are finitely many r , this is at least am^κ for $m \gg 0$.

For the upper bound, first suppose $H^0(X, L) \neq 0$, and write $L = \mathcal{O}_X(D)$ with D effective. On the nonsingular X , decompose

$$D = D_{\text{vert}} + D_{\text{horiz}},$$

where each component of D_{vert} maps to a proper subvariety of Y and each component of D_{horiz} dominates Y . These are Cartier divisors since X is nonsingular. We assert that, under the usual realization by rational functions,

$$H^0(X, \mathcal{O}_X(mD_{\text{vert}})) = H^0(X, \mathcal{O}_X(mD)) \quad (m \geq 1). \quad (3.1)$$



Granting the assertion, choose an ample Cartier divisor H on Y with $D_{\text{vert}} \leq \varphi^* H$. Then

$$\begin{aligned} h^0(X, \mathcal{O}_X(mD)) &= h^0(X, \mathcal{O}_X(mD_{\text{vert}})) \\ &\leq h^0(X, \varphi^* \mathcal{O}_Y(mH)) = h^0(Y, \mathcal{O}_Y(mH)) \\ &= \frac{(H^\kappa)}{\kappa!} m^\kappa + O(m^{\kappa-1}) \leq Am^\kappa \end{aligned}$$

for $m \gg 0$. To prove (3.1), one inclusion is immediate from effectivity. For the other, every section s of $\mathcal{O}_X(mD)$ must vanish along mD_{horiz} . Indeed, its zero divisor is $(s)_0 = \text{div}(s) + mD$, and the required inequality is

$$\text{div}(s) + mD_{\text{vert}} = (s)_0 - mD_{\text{horiz}} \geq 0.$$

If a section failed this inequality at a horizontal prime divisor, its restriction to a very general fibre would have a different zero divisor from the restriction of the canonical section of mD . The fibre avoids D_{vert} , meets this horizontal divisor, and can be chosen to avoid the other exceptional intersection conditions. These two restricted sections would be linearly independent. This contradicts $h^0(F, \mathcal{O}_F(mD)) \leq 1$, proved in Step 3 of the theorem.

For the construction of H , write $D_{\text{vert}} = \sum_i a_i \Gamma_i$ and $Z_i = \varphi(\Gamma_i) \subsetneq Y$. Choose a very ample divisor A_Y on Y . For $m_i \gg 0$, $\mathcal{I}_{Z_i} \otimes \mathcal{O}_Y(m_i A_Y)$ is globally generated and has a non-zero section s_i . Put $H_i = (s_i)_0$, so that $Z_i \subseteq \text{Supp } H_i$ and

$$\varphi^* H_i = b_i \Gamma_i + E_i, \quad b_i \geq 1, \quad E_i \geq 0.$$

Choose positive integers c_i with $c_i b_i \geq a_i$. Then $H = \sum_i c_i H_i$ is ample and

$$\varphi^* H - D_{\text{vert}} = \sum_i (c_i b_i - a_i) \Gamma_i + \sum_i c_i E_i \geq 0.$$

If $D_{\text{vert}} = 0$, simply choose any effective ample H .

Finally remove the assumption $H^0(X, L) \neq 0$. Choose $p > 0$ in $N(L)$ and apply the upper-bound argument to $L^{\otimes p}$, whose Iitaka fibration and dimension are the same. For each residue $r \in \{0, e, \dots, p-e\}$ choose $b_r \in N(L)$ with $b_r \equiv -r \pmod{p}$. Such integers exist by the semigroup property. If $m \in N(L)$ and $m \equiv r \pmod{p}$, multiplication by a non-zero section of $L^{\otimes b_r}$ gives

$$h^0(X, L^{\otimes m}) \leq h^0(X, L^{\otimes(m+b_r)}) \leq C \left(\frac{m+b_r}{p} \right)^\kappa \leq A' m^\kappa$$

for $m \gg 0$. There are only finitely many b_r , so a uniform A' works. $\square$

Remark 3.1.36. For a normal projective variety and $\kappa = \kappa(X, L) \geq 0$, it is now known that

$$\lim_{\substack{m \rightarrow \infty \\ m \in N(X, L)}} \frac{h^0(X, L^{\otimes m})}{m^\kappa}$$

exists and is positive and finite. This follows from the limit theorem for graded linear series of Okounkov, Lazarsfeld–Mustaă, and Kaveh–Khovanskii; see [Cut12, Theorem 3.1]. Indeed, that theorem gives the limit along multiples of the index $e(L)$, and these are exactly the sufficiently large elements of $N(L)$. Without this restriction, zeros in the other residue classes can prevent a limit when $e(L) > 1$. In the big case the index is 1, as we will see in Corollary 3.2.10; the full sequence therefore has a limit.

Remark 3.1.37. For a normal projective variety, the preceding corollary gives an asymptotic definition of $\kappa(X, L)$: if every positive power has no non-zero section, set $\kappa(X, L) = -\infty$; otherwise it is the largest integer $d \geq 0$ such that

$$\limsup_{m \rightarrow \infty} \frac{h^0(X, L^{\otimes m})}{m^d} > 0.$$

One can restrict to $m \in N(L)$ without changing this limsup. This is the asymptotic definition used in the lecture referred to at the beginning of the chapter. For non-normal X , apply it to ν^*L on the normalization to retain Definition 3.1.3.

Proof. Write κ_{class} and κ_{def} for the two definitions. *Case 1.* If $h^0(X, L^{\otimes m}) = 0$ for every $m > 0$, no non-negative integer satisfies the inequality, and both definitions give $-\infty$. *Case 2.* Otherwise, Corollary 3.1.35 gives

$$\frac{h^0(X, L^{\otimes m})}{m^{\kappa_{\text{class}}}} \geq a > 0$$

for all sufficiently large $m \in N(L)$, so $\kappa_{\text{def}} \geq \kappa_{\text{class}}$. If $d > \kappa_{\text{class}}$, the upper bound gives

$$0 \leq \frac{h^0(X, L^{\otimes m})}{m^d} \leq Am^{\kappa_{\text{class}}-d} \rightarrow 0.$$

Thus no such d can occur, proving equality of the definitions. $\square$

3.2 Big Line Bundles and Cartier Divisors

3.2A Basic Properties of Big Cartier Divisors

Definition 3.2.1 (Big). A line bundle L on a projective variety X is big if $\kappa(X, L) = \dim X$. A Cartier divisor D is big if $\mathcal{O}_X(D)$ is so.

Example 3.2.2. The pullback of an ample line bundle under a dominant generically finite morphism $f : X \rightarrow Y$ of projective varieties is big.

Proof. We use “generically finite” for a dominant map with finite extension of function fields. Passing to the normalization of X , we may assume X normal. Put $\dim X = \dim Y = n$ and let A be ample on Y . Dominance gives $\mathcal{O}_Y \hookrightarrow f_*\mathcal{O}_X$, hence

$$A^{\otimes m} \hookrightarrow A^{\otimes m} \otimes f_*\mathcal{O}_X \cong f_*(f^*A^{\otimes m}).$$

Consequently

$$\begin{aligned} h^0(X, f^*A^{\otimes m}) &\geq h^0(Y, A^{\otimes m}) \\ &= \frac{m^n}{n!} \int_Y c_1(A)^n + O(m^{n-1}) \geq Cm^n \end{aligned}$$

for some $C > 0$ and $m \gg 0$. The growth characterization of Iitaka dimension in Remark 3.1.37 proves bigness. The zero-dimensional case is immediate. $\square$

Example 3.2.3. If X is normal, L is big if and only if, for some $m > 0$, the rational map $\varphi_m : X \dashrightarrow Y_m$ is birational onto its image.

Proof. One direction is immediate from the definition. Conversely, for a big line bundle the Iitaka fibration has $\dim X_\infty = \dim Y_\infty$. The field extension $\mathbb{C}(Y_\infty) \subseteq \mathbb{C}(X_\infty)$ is finite and the first field is algebraically closed in the second, so they are equal. The Iitaka fibration, and hence φ_m for sufficiently large $m \in N(L)$, is birational. If $\dim X = 0$, both sides are automatic. $\square$

Although the definition uses normalization, in the big case we can measure growth on the original variety. The following criterion is often used as the definition; see [KM98, 2.59].

Lemma 3.2.4. *Let X be a projective variety of dimension n , not necessarily normal. A line bundle L is big if and only if there is $C > 0$ such that*

$$h^0(X, L^{\otimes m}) \geq Cm^n$$

for all sufficiently large and divisible m .

Proof. Let $\nu : X' \rightarrow X$ be the normalization and $L' = \nu^*L$. The exact sequence

$$0 \rightarrow \mathcal{O}_X \rightarrow \nu_* \mathcal{O}_{X'} \rightarrow \eta \rightarrow 0$$

has η supported in dimension at most $n - 1$. By the projection formula,

$$\begin{aligned} h^0(X, L^{\otimes m}) &\leq h^0(X', L'^{\otimes m}) \\ &\leq h^0(X, L^{\otimes m}) + h^0(X, \eta \otimes L^{\otimes m}). \end{aligned}$$

The last term is $O(m^{n-1})$ by the cohomology growth bound of Chapter 1. Thus the two spaces have the same leading growth in degree n . If L is big, apply Corollary 3.1.35 on X' along sufficiently large multiples of $e(L')$ and absorb the lower-order error to obtain the stated bound on X . Conversely, such a lower bound on X also holds on X' , forcing $\kappa(X', L') = n$ by Corollary 3.1.35 and the general $O(m^n)$ bound. For $n = 0$ the normalization is an isomorphism and the assertion is immediate. $\square$

Example 3.2.5 (Blow-up of projective space). Let $S \subseteq \mathbb{P}^n$ be a finite reduced set of s points, with ideal sheaf $\mathcal{I}$, and let $\mu : X = \text{Bl}_S \mathbb{P}^n \rightarrow \mathbb{P}^n$. Write $E = E_S$ for the divisor defined by the inverse-image ideal and $H = H_S$ for the pullback of a hyperplane. For positive integers d, r put $D = dH - rE$. Then

$$\begin{aligned} \mu_* \mathcal{O}_X(mD) &= \mu_*(\mathcal{O}_X(mdH) \otimes \mathcal{I}_E^{mr}) \\ &= \mathcal{O}_{\mathbb{P}^n}(md) \otimes \mathcal{I}^{mr}. \end{aligned}$$

Here $\mathcal{I}^{mr}$ is the ordinary ideal power, rather than its tensor power; for the blow-up of distinct smooth points, $\mu_* \mathcal{O}_X(-mrE) = \mathcal{I}^{mr}$. Thus $H^0(X, \mathcal{O}_X(mD))$ is the

space of degree md hypersurfaces vanishing to order at least mr at each point of S . We have

$$h^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(md)) = \binom{n+md}{n} = \frac{d^n}{n!} m^n + O(m^{n-1}).$$

Vanishing to order mr at one smooth point imposes at most

$$\binom{mr-1+n}{n} = \frac{r^n}{n!} m^n + O(m^{n-1})$$

linear conditions, obtained by setting all Taylor coefficients of degree less than mr to zero. No independence of these conditions at different points is needed for the inequality

$$h^0(X, \mathcal{O}_X(mD)) \geq \frac{d^n - sr^n}{n!} m^n + O(m^{n-1}).$$

Therefore D is big whenever $d^n - sr^n > 0$.

The first essential result is due to Kodaira.

Proposition 3.2.6 (Kodaira's lemma). *Let D be a big Cartier divisor on a projective variety X , and F any effective Cartier divisor. Then*

$$H^0(X, \mathcal{O}_X(mD - F)) \neq 0$$

for all sufficiently large and divisible m .

Proof. Let $\dim X = n$. From

$$0 \longrightarrow \mathcal{O}_X(mD - F) \longrightarrow \mathcal{O}_X(mD) \longrightarrow \mathcal{O}_F(mD) \longrightarrow 0$$

we compare $h^0(X, mD) \geq Cm^n$ with $h^0(F, \mathcal{O}_F(mD)) = O(m^{n-1})$. For sufficiently large and divisible m the first is larger, so the kernel on sections is non-zero. If $F = 0$ the assertion is immediate. $\square$

Kodaira's lemma gives the following characterization.

Corollary 3.2.7 (Characterization of big Cartier divisors). *For a Cartier divisor D on a projective variety X , the following are equivalent:*

- (i) D is big.
- (ii) For any ample integral Cartier divisor A , there are $m > 0$ and an effective Cartier divisor N with $mD \sim_{\text{lin}} A + N$.
- (iii) The same statement holds for some ample Cartier divisor A .
- (iv) There are an ample Cartier divisor A , an integer $m > 0$ and an effective Cartier divisor N with $mD \equiv_{\text{num}} A + N$.

Proof. “(i) $\Rightarrow$ (ii)”: Choose $r \gg 0$ such that $rA \sim_{\text{lin}} H_r$ and $(r+1)A \sim_{\text{lin}} H_{r+1}$ are effective; Serre’s global generation theorem gives this choice. Apply Kodaira’s lemma with $F = H_{r+1}$ to get

$$mD \sim_{\text{lin}} H_{r+1} + N' \sim_{\text{lin}} A + (H_r + N'), \quad N' \geq 0.$$

Take $N = H_r + N'$. The implications (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) are immediate. For (iv) $\Rightarrow$ (i), the Cartier divisor $mD - N$ is numerically equivalent to A , hence ample by Nakai–Moishezon. Choose $t > 0$ such that $H = t(mD - N)$ is very ample. Then $tmD \sim_{\text{lin}} H + tN$. Multiplication by the section of ktN embeds the sections of kH in those of $ktmD$. The growth criterion proves D big. $\square$

Corollary 3.2.8 (Numerical nature of bigness). *Bigness of a Cartier divisor depends only on its numerical equivalence class.*

Example 3.2.9 (Cohomological characterization of big Cartier divisors). A Cartier divisor D on a projective variety X is big if and only if, for every coherent sheaf $\mathcal{F}$, there is an integer $m = m(\mathcal{F}) > 0$ such that $\mathcal{F} \otimes \mathcal{O}_X(mD)$ is generated by its global sections on some non-empty open subset of X .

Proof. “ $\Rightarrow$ ”: Choose an ample Cartier divisor A such that $\mathcal{F}(A)$ is globally generated. Write $mD \sim_{\text{lin}} A + N$ with $N \geq 0$ by Corollary 3.2.7. Multiplying generators of $\mathcal{F}(A)$ by the canonical section of N generates $\mathcal{F}(mD)$ on $X \setminus \text{Supp } N$. “ $\Leftarrow$ ”: Take $\mathcal{F} = \mathcal{O}_X(-A)$ for a very ample A . Generation on a non-empty open set gives a non-zero section of $\mathcal{O}_X(mD - A)$. Thus $mD \sim_{\text{lin}} A + N$ with $N \geq 0$, and D is big by Corollary 3.2.7. $\square$

Concerning the exponent of a big divisor, we obtain:

Corollary 3.2.10 (Exponent of big Cartier divisors). *If D is a big Cartier divisor on a projective variety, then $e(D) = 1$. Consequently all sufficiently large multiples of D are effective up to linear equivalence.*

Proof. Choose a very ample H such that $H - D \sim_{\text{lin}} H_1$ is effective. Corollary 3.2.7 and its proof give $m \geq 2$ and $N \geq 0$ with $mD \sim_{\text{lin}} H + N$. Then

$$(m-1)D \sim_{\text{lin}} (H - D) + N \sim_{\text{lin}} H_1 + N$$

is effective as well. The two consecutive integers $m, m-1$ belong to $N(D)$, so $e(D) = 1$. The final assertion is the semigroup fact of Definition 3.1.1. $\square$

Combining this with normalization in Lemma 3.2.4 shows that its growth bound, and the non-vanishing in Kodaira’s lemma, in fact hold for every sufficiently large integer m when D is big.

Corollary 3.2.11 (Restrictions of big Cartier divisors). *Let L be a big line bundle on a projective variety X . There is a proper closed subset $V \subsetneq X$ such that, for every subvariety $Y \subseteq X$ not contained in V , $L|_Y$ is big. In particular, restriction to a general member of a very ample linear series is big (componentwise if that member has dimension zero).*

Proof. Write $L = \mathcal{O}_X(D)$ and $mD \sim_{\text{lin}} H + N$ with H very ample and N effective. If $Y \not\subseteq \text{Supp } N$, restriction gives the sum of a very ample divisor and an effective Cartier divisor on Y . Apply Corollary 3.2.7 again, and take $V = \text{Supp } N$. $\square$

Example 3.2.12 (Restriction fails to preserve bigness). On $X = \text{Bl}_p \mathbb{P}^2$, let E be exceptional and H the pullback of a line. The bundle $\mathcal{O}_X(H + E)$ is big, but $\mathcal{O}_X(H + E)|_E \cong \mathcal{O}_{\mathbb{P}^1}(-1)$ is not big.

An integral Cartier divisor is big if and only if some, or every, positive multiple is big. This leads to the following definition.

Definition 3.2.13 (Big $\mathbb{Q}$ -Cartier divisors). A $\mathbb{Q}$ -Cartier divisor D is big if mD is integral Cartier and big for some positive integer m .

By passing to integral multiples, bigness is a numerical property for $\mathbb{Q}$ -Cartier divisors. We now give Siu's numerical criterion.

Theorem 3.2.14 (Numerical criterion for bigness). *Let X be a projective variety of dimension $n \geq 1$, and D, E nef $\mathbb{Q}$ -Cartier divisors. If*

$$(D^n) > n(D^{n-1} \cdot E),$$

then $D - E$ is big.

Proof. Fix an ample Cartier divisor H . The polynomial

$$f(t) = ((D + tH)^n) - n((D + tH)^{n-1} \cdot (E + tH))$$

is positive at 0, hence remains positive for small $t > 0$. Replace D, E by $D + \varepsilon H, E + \varepsilon H$ with small positive rational ε . Both become ample, and their difference is *unchanged*. Multiplying both by a common positive integer, we may assume they are integral and very ample. Choose general divisors $E_1, E_2, \dots \in |E|$ successively so that the local equation of E_j is a non-zero-divisor modulo the sum of the previous divisors. Equivalently, avoid the finitely many associated points of the previous Cartier thickening. Then intersection of their ideal sheaves is their product, and the sections vanishing on all E_i , $1 \leq i \leq m$, are exactly those vanishing on $\sum_{i=1}^m E_i \sim mE$. Thus there is an exact sequence at the first two terms

$$0 \longrightarrow H^0(X, m(D - E)) \longrightarrow H^0(X, mD) \longrightarrow \bigoplus_{i=1}^m H^0(E_i, \mathcal{O}_{E_i}(mD)).$$

For $m \gg 0$, the dimensions on the right are independent of i . Indeed, the sequence for $E_i \sim E$ and Serre vanishing for $\mathcal{O}_X(mD - E)$ express them as $h^0(X, mD) - h^0(X, mD - E)$, uniformly for every E_i . Riemann–Roch gives

$$\begin{aligned} h^0(X, m(D - E)) &\geq h^0(X, mD) - \sum_{i=1}^m h^0(E_i, \mathcal{O}_{E_i}(mD)) \\ &\geq \frac{(D^n) - n(D^{n-1} \cdot E)}{n!} m^n + O(m^{n-1}). \end{aligned}$$

Here the restriction Hilbert polynomial has leading term $(D^{n-1} \cdot E)m^{n-1}/(n-1)!$; for $n = 1$ it is the constant length (E) . The leading coefficient is positive, proving bigness. $\square$

Theorem 3.2.15 (Bigness of nef Cartier divisors). *A nef Cartier divisor D on a projective variety of dimension n is big if and only if $(D^n) > 0$.*

Proof. “ $\Leftarrow$ ”: For $n \geq 1$, apply Theorem 3.2.14 with $E = 0$. The case $n = 0$ is immediate. “ $\Rightarrow$ ”: Write $mD \sim_{\text{lin}} H + N$, with H very ample and $N \geq 0$. Since D is nef, $(D^{n-1} \cdot N) \geq 0$, hence

$$m(D^n) = (D^{n-1} \cdot (H + N)) \geq (D^{n-1} \cdot H).$$

We proceed by induction. For a curve, the same decomposition gives $m \deg D = \deg H + \deg N > 0$, proving the base case directly. For $n > 1$, choose a general integral member of $|H|$. By Corollary 3.2.11, $D|_H$ is big, and it is also nef. The induction hypothesis gives $(D^{n-1} \cdot H) > 0$, so $(D^n) > 0$. $\square$

Remark 3.2.16 (Alternative approach to Theorem 3.2.15). The asymptotic result for nef divisors deduced from Fujita vanishing in Corollary 1.4.46 gives

$$h^0(X, mD) = \frac{(D^n)}{n!} m^n + O(m^{n-1}).$$

Thus Lemma 3.2.4 gives the same characterization directly.

Example 3.2.17 (Big line bundles with negative numerics). Without nefness, intersection numbers of big line bundles can be negative. Let $X = \text{Bl}_p \mathbb{P}^2$, with exceptional divisor E and a very ample divisor A . Then $D_l = A + lE$ is big for every integer $l \geq 0$, but

$$(D_l^2) = (A^2) + 2l(A \cdot E) - l^2 \longrightarrow -\infty.$$

Example 3.2.18 (Alternative characterization of big and nef Cartier divisors). A Cartier divisor D on a projective variety is nef and big if and only if there is an effective Cartier divisor N such that $D - \frac{1}{k}N$ is ample for all $k \gg 0$.

Proof. “ $\Leftarrow$ ”: Bigness follows from the ample-plus-effective decomposition. For every curve C , $(D \cdot C)$ is the limit of the positive numbers $((D - \frac{1}{k}N) \cdot C)$, so D is nef. “ $\Rightarrow$ ”: Write $mD \equiv_{\text{num}} A + N$, with A ample and $N \geq 0$. For $k > m$,

$$kD - N \equiv_{\text{num}} (k - m)D + A$$

is the sum of a nef and an ample divisor, hence ample. $\square$

3.2B Pseudo-effective and Big Cones

We now rephrase Kodaira’s lemma in the language of numerical geometry. As before, X is a projective variety.

Definition 3.2.19 (Big $\mathbb{R}$ -Cartier divisors). An $\mathbb{R}$ -Cartier divisor $D \in \text{Div}_{\mathbb{R}}(X)$ is big if it can be written $D = \sum_i a_i D_i$, where each D_i is a big integral Cartier divisor and $a_i > 0$ is real.

The definition is stable under addition and positive scalar multiplication.

Proposition 3.2.20 (Formal properties of big $\mathbb{R}$ -Cartier divisors). *Let D, D' be $\mathbb{R}$ -Cartier divisors on X .*

- (1) *If $D \equiv_{\text{num}} D'$, then D is big if and only if D' is big.*
- (2) *D is big if and only if $D \equiv_{\text{num}} A + N$, where A is ample and N is an effective $\mathbb{R}$ -Cartier divisor.*

Proof. (1) For rational divisors, clear denominators and use Corollary 3.2.8. For real divisors one more argument is needed. As in Proposition 1.3.14, numerical relations in the span of finitely many integral Cartier divisors are defined over $\mathbb{Q}$. Thus a numerically trivial real Cartier divisor can be written as $P = \sum_j b_j P_j$ with P_j numerically trivial integral Cartier divisors. If B is big and integral, choose rational numbers on either side of each b_j . Expressing $(b_j)_j$ as a convex combination of the rational vertices of this box expresses $B + P$ as a convex combination of rational divisors numerically equivalent to B . They are big, so $B + P$ is big. If $D = \sum_i a_i D_i$ is big, apply the same argument to $D_1 + P/a_1$ and leave the other summands unchanged. This proves invariance under adding P , and interchanging D, D' proves (1).

(2) If D is big, apply Corollary 3.2.7 to each integral summand and take the same positive real combination of the ample and effective parts. Conversely, an ample real divisor is a positive combination of ample, hence big, integral divisors. To add an effective real divisor it suffices, one summand at a time, to treat $B + sN$, with B big integral, N effective integral and $s > 0$ real. For rational s this is Corollary 3.2.7. In general choose $0 < s_1 < s < s_2$ rational and $t \in (0, 1)$ with $s = ts_1 + (1 - t)s_2$. Then

$$B + sN = t(B + s_1N) + (1 - t)(B + s_2N)$$

is a positive real combination of big rational divisors. Finally use (1) to pass from $A + N$ to the numerically equivalent divisor D . $\square$

Example 3.2.21 (Big and nef $\mathbb{R}$ -Cartier divisors). If D is a nef and big real Cartier divisor, there is an effective real Cartier divisor N such that $D - \frac{1}{k}N$ is ample for every sufficiently large positive integer k .

Proof. Write $D \equiv_{\text{num}} A + N$ by Proposition 3.2.20. For $k > 1$,

$$D - \frac{1}{k}N \equiv_{\text{num}} \left(1 - \frac{1}{k}\right) D + \frac{1}{k}A$$

is the sum of a nef real divisor and an ample real divisor, hence ample. $\square$

Corollary 3.2.22 (Openness of big Cartier divisors). *If D is a big real Cartier divisor and $E_1, \dots, E_t$ are arbitrary real Cartier divisors, then $D + \varepsilon_1 E_1 + \dots + \varepsilon_t E_t$ remains big for all sufficiently small real $|\varepsilon_i|$.*

Proof. Write $D \equiv_{\text{num}} A + N$ with A ample and N effective. The divisor $A + \sum_i \varepsilon_i E_i$ remains ample by openness of ampleness. Apply Proposition 3.2.20(2). $\square$

Numerical invariance allows us to speak of big real classes in $N^1(X)_{\mathbb{R}}$.

Definition 3.2.23 (Big and pseudo-effective cones). The big cone $\text{Big}(X) \subseteq N^1(X)_{\mathbb{R}}$ is the convex cone of all big real Cartier divisor classes. The pseudo-effective cone $\overline{\text{Eff}}(X) \subseteq N^1(X)_{\mathbb{R}}$ is the closure of the convex cone generated by effective real Cartier divisor classes. A real Cartier divisor is pseudo-effective if its class lies in $\overline{\text{Eff}}(X)$.

Theorem 3.2.24. *The big cone is the interior of the pseudo-effective cone, and the pseudo-effective cone is the closure of the big cone:*

$$\text{Big}(X) = \text{int}(\overline{\text{Eff}}(X)), \quad \overline{\text{Eff}}(X) = \overline{\text{Big}(X)}.$$

Proof. The big cone is open, and every big class is the sum of an ample class and an effective class. Ample classes are positive real combinations of classes with effective positive multiples, so $\text{Big}(X) \subseteq \overline{\text{Eff}}(X)$. Taking closures gives one inclusion. Conversely, if $\eta \in \overline{\text{Eff}}(X)$, choose effective *real* classes $\eta_k \rightarrow \eta$ and fix an ample class α . Then

$$\eta = \lim_{k \rightarrow \infty} (\eta_k + k^{-1}\alpha),$$

and each class on the right is big. This proves equality of closures. Finally, a non-empty open convex set in a finite-dimensional real vector space equals the interior of its closure. For example, a separating hyperplane at any point outside the open convex set shows that the point cannot be interior to its closure. Apply this to $\text{Big}(X)$. $\square$

Remark 3.2.25 (Pseudo-effective Cartier divisors on a surface). On a smooth projective surface, divisors and curves are identified via the intersection pairing. Under this identification $\overline{\text{Eff}}(X) = \overline{\text{NE}}(X)$ and

$$\overline{\text{Eff}}(X) = \text{Nef}(X)^*.$$

Thus D is pseudo-effective if and only if $(D \cdot H) \geq 0$ for every nef divisor H . This says that D lies in the *dual of the nef cone*; it does not assert that D itself is nef.

3.2C Volume of a Big Cartier Divisor

There is an interesting invariant measuring the asymptotic growth of $|mD|$ as $m \rightarrow \infty$.

Definition 3.2.26 (Volume of a line bundle). For a line bundle L on an n -dimensional projective variety X , define

$$\text{vol}(L) = \text{vol}_X(L) = \limsup_{m \rightarrow \infty} \frac{h^0(X, L^{\otimes m})}{m^n/n!}.$$

The volume $\text{vol}(D) = \text{vol}_X(D)$ of a Cartier divisor is defined via $\mathcal{O}_X(D)$.

The cohomology growth bound makes volume finite, and $\text{vol}(L) > 0$ if and only if L is big. If L is nef, Fujita vanishing and asymptotic Riemann–Roch give

$$h^0(X, L^{\otimes m}) = \frac{\int_X c_1(L)^n}{n!} m^n + O(m^{n-1}), \quad \text{vol}(L) = \int_X c_1(L)^n.$$

For a zero-dimensional variety all these bundles are trivial and the volume is 1. The proofs below involving hyperplane divisors are given for $n \geq 1$; the zero-dimensional assertions are immediate.

Example 3.2.27 (Volume on a blow-up of projective space). Blow up $\mathbb{P}^n$ at s distinct points and put $D = dH - rE$, as in Example 3.2.5. The computation there gives

$$h^0(X, mD) \geq \frac{d^n - sr^n}{n!} m^n + O(m^{n-1}), \quad \text{vol}(D) \geq d^n - sr^n.$$

Proposition 3.2.28 (Properties of the volume). *Let D be a big Cartier divisor on an n -dimensional projective variety.*

- (i) *For every positive integer a , $\text{vol}(aD) = a^n \text{vol}(D)$.*
- (ii) *Given any Cartier divisor N and $\varepsilon > 0$, there is an integer $p_0 = p_0(D, N, \varepsilon)$ such that*

$$\frac{|\text{vol}(pD - N) - \text{vol}(pD)|}{p^n} < \varepsilon \quad (p > p_0).$$

We begin with two lemmas.

Lemma 3.2.29. *Let L be a line bundle and A a very ample divisor on X , with $n \geq 1$. For very general $E, E' \in |A|$, one has*

$$h^0(E, L^{\otimes m}|_E) = h^0(E', L^{\otimes m}|_{E'}) \quad \text{for all } m \geq 1,$$

and hence $\text{vol}_E(L|_E) = \text{vol}_{E'}(L|_{E'})$. For $n = 1$, the volume on the zero-dimensional divisor means its length; the same assertion holds.

Proof. Let $V = H^0(X, \mathcal{O}_X(A))$ and let $P = |A|$ be the space of lines in V . With our quotient convention this is $P = \mathbb{P}(V^\vee)$. Choose a basis $s_0, \dots, s_N$ of V and dual homogeneous coordinates $y_0, \dots, y_N$ on P . On $X \times P$ the line bundle

$$\mathcal{H} = \text{pr}_1^* \mathcal{O}_X(A) \otimes \text{pr}_2^* \mathcal{O}_P(1)$$

has the universal section $s = \sum_i s_i \boxtimes y_i$. Its zero divisor $\mathcal{D} \subseteq X \times P$ has fibre over $t = [c_0 : \dots : c_N]$ equal to the divisor of $\sum_i c_i s_i$, namely $E_t \in |A|$. We have the two projections

$$X \xleftarrow{p_1} \mathcal{D} \subseteq X \times P \xrightarrow{p_2} P = |A|$$

The exact sequence

$$0 \longrightarrow \mathcal{H}^{-1} \xrightarrow{\cdot s} \mathcal{O}_{X \times P} \longrightarrow \mathcal{O}_{\mathcal{D}} \longrightarrow 0$$

shows that $\mathcal{D} \rightarrow P$ is flat. Indeed, the first two sheaves are flat over P , and on every fibre multiplication by the corresponding non-zero section is injective, since X is geometrically integral. The local criterion for flatness then makes the quotient flat over P . For each $m \geq 1$, the sheaf $(\text{pr}_1^* L^{\otimes m})|_{\mathcal{D}}$ is also flat over P . Properness and upper semicontinuity give a dense open $U_m \subseteq P$ on which $h^0(E_t, L^{\otimes m}|_{E_t})$ is constant. Choose E, E' in the intersection of all U_m . For $n \geq 2$, also impose Bertini's open condition that they be integral; for $n = 1$ they are finite divisors and their section dimensions equal their lengths. The equality of all section dimensions proves the equality of volumes. Notice that bigness of L was not needed in this proof. $\square$

Lemma 3.2.30. *For any Cartier divisor D and any fixed integer $a > 0$,*

$$\limsup_{m \rightarrow \infty} \frac{h^0(X, mD)}{m^n/n!} = \limsup_{k \rightarrow \infty} \frac{h^0(X, akD)}{(ak)^n/n!}.$$

Proof. If D is not big, its volume is zero, and both sides vanish. Assume D big, so every sufficiently large integer belongs to $N(D)$. For $r \geq 0$ put

$$v_r = \limsup_{k \rightarrow \infty} \frac{h^0(X, (ak + r)D)}{(ak + r)^n/n!}.$$

For any fixed r_0 , division with a shifted remainder set partitions the sequence into the a subsequences with $r \in \{r_0 + 1, \dots, r_0 + a\}$. Hence $\text{vol}(D) = \max\{v_{r_0+1}, \dots, v_{r_0+a}\}$. Choose r_0 so large that $h^0(X, rD) \neq 0$ for $r \geq r_0$, and then choose q with $qa - (r_0 + a) \geq r_0$. For each of these remainders choose

$$D_r \in |rD|, \quad D'_r \in |(qa - r)D|.$$

Multiplication gives

$$\mathcal{O}_X(kaD) \hookrightarrow \mathcal{O}_X((ka + r)D) \hookrightarrow \mathcal{O}_X((k + q)aD),$$

and therefore

$$h^0(X, kaD) \leq h^0(X, (ka + r)D) \leq h^0(X, (k + q)aD).$$

After division by $(ka)^n/n!$ and taking limsup, both outer expressions have limsup v_0 : the fixed shifts change the denominators by factors tending to 1. Thus $v_0 \leq v_r \leq v_0$, proving the assertion. $\square$

Proof of Proposition 3.2.28. By Lemma 3.2.30,

$$\text{vol}(aD) = \limsup_k \frac{h^0(X, akD)}{k^n/n!} = a^n \limsup_k \frac{h^0(X, akD)}{(ak)^n/n!} = a^n \text{vol}(D).$$

The same argument proves (i) for every Cartier divisor, including non-big ones. For (ii), write $N \sim_{\text{lin}} A - B$ with A, B very ample effective divisors. Since D is big, choose $r > 0$ and $B_1 \geq 0$ with $rD - B \sim_{\text{lin}} B_1$. Then

$$pD - N \sim_{\text{lin}} (p + r)D - (A + B_1).$$

Since $(p + r)^n/p^n \rightarrow 1$, it suffices to establish $\text{vol}(qD - F)/q^n \rightarrow \text{vol}(D)$ for the effective divisor $F = A + B_1$. Thus we may assume N effective. For a second effective divisor N' , multiplication of sections gives

$$\text{vol}(pD - (N + N')) \leq \text{vol}(pD - N) \leq \text{vol}(pD).$$

Choose N' so that $N + N'$ is very ample. It is enough to prove (ii) for this larger divisor, so we reduce to N very ample.

Choose distinct very general divisors $E_1, E_2, \dots \in |N|$, with the proper-intersection condition used in the proof of Theorem 3.2.14. By Lemma 3.2.29 they have the same section dimensions in all powers of $\mathcal{O}_X(D)$ as a fixed very general $E \in |N|$. For each $m, p > 0$, the exact restriction sequence on sections gives

$$\begin{aligned} h^0(X, m(pD - N)) &\geq h^0(X, mpD) - \sum_{i=1}^m h^0(E_i, \mathcal{O}_{E_i}(mpD)) \\ &= h^0(X, mpD) - mh^0(E, \mathcal{O}_E(mpD)). \end{aligned}$$

The divisors E_i remain distinct in this sequence; only their section *dimensions* are replaced by the same number. Divide by $m^n/n!$. Take a subsequence on which the first term on the right approaches $\text{vol}_X(pD)$, and bound the limsup of the subtracted term by $n \text{vol}_E(pD|_E)$. This gives

$$\text{vol}_X(pD - N) \geq \text{vol}_X(pD) - n \text{vol}_E(pD|_E).$$

Homogeneity in dimension $n - 1$ gives $\text{vol}_E(pD|_E) = p^{n-1} \text{vol}_E(D|_E)$, including the length interpretation when $n = 1$. Consequently

$$\text{vol}(D) \geq \frac{\text{vol}(pD - N)}{p^n} \geq \text{vol}(D) - \frac{n}{p} \text{vol}_E(D|_E),$$

and (ii) follows by letting $p \rightarrow \infty$. $\square$

Remark 3.2.31. The second statement hints at the continuity properties of volume.

Remark 3.2.32 (Volume of a $\mathbb{Q}$ -Cartier divisor). For a $\mathbb{Q}$ -Cartier divisor D , choose $a > 0$ such that aD is integral Cartier and define $\text{vol}(D) = a^{-n} \text{vol}(aD)$. Homogeneity shows independence of a . Equivalently, take the defining limsup over the integers m for which mD is integral Cartier.

Example 3.2.33 (Volume of a difference of nef Cartier divisors, I). For nef $\mathbb{Q}$ -Cartier divisors D, E ,

$$\text{vol}(D - E) \geq (D^n) - n(D^{n-1} \cdot E).$$

Indeed, when D, E are ample this is the section estimate in the proof of Theorem 3.2.14, followed by clearing denominators and homogeneity. In the nef case apply that estimate to $D + \varepsilon H, E + \varepsilon H$ with $\varepsilon > 0$ rational. Their difference is $D - E$, and letting $\varepsilon \rightarrow 0$ on the intersection-polynomial side proves the result. No continuity theorem for volume is needed at this stage.

Using Fujita vanishing, we next deduce numerical invariance of volume.

Proposition 3.2.34 (Numerical nature of volume). *If D, D' are numerically equivalent $\mathbb{Q}$ -Cartier divisors on X , then $\text{vol}(D) = \text{vol}(D')$. Thus $\text{vol}(\xi)$ is defined for $\xi \in N^1(X)_{\mathbb{Q}}$.*

Lemma 3.2.35. *There is a fixed Cartier divisor N such that*

$$H^0(X, \mathcal{O}_X(N + P)) \neq 0$$

for every numerically trivial Cartier divisor P . In particular, $N \pm mP_0$ is linearly equivalent to an effective divisor for every numerically trivial P_0 and every $m \in \mathbb{Z}$.

Proof. Fix a very ample divisor B . By Fujita vanishing there is a large multiple F of B such that $H^i(X, \mathcal{O}_X(F + A)) = 0$ for $i > 0$ and every nef Cartier divisor A . For $1 \leq i \leq n$, use $A = P + (n - i)B$. Castelnuovo–Mumford regularity shows that $F + P + nB$ is globally generated. Thus $N = F + nB$ works for all P . Applying this to $\pm mP_0$ proves the last assertion. $\square$

Proof of Proposition 3.2.34. Clear denominators to assume the divisors integral. By numerical invariance of bigness, either both volumes are zero or both divisors are big. In the latter case write $D' = D + P$, with P numerically trivial. Fix the divisor N of Lemma 3.2.35, independent of p . A non-zero section of $\mathcal{O}_X(N - pP)$ gives, for every $m > 0$, an inclusion

$$\mathcal{O}_X(m(p(D + P) - N)) \hookrightarrow \mathcal{O}_X(mpD).$$

It follows that $\text{vol}(p(D + P) - N) \leq p^n \text{vol}(D)$. By Proposition 3.2.28(ii), division by p^n and $p \rightarrow \infty$ give $\text{vol}(D + P) \leq \text{vol}(D)$. Replace D by $D + P$ and P by $-P$ for the opposite inequality. $\square$

Proposition 3.2.36 (Birational invariance of volume, I). *Let $\nu : X' \rightarrow X$ be a birational morphism of projective varieties. For an integral or $\mathbb{Q}$ -Cartier divisor D , put $D' = \nu^*D$. Then $\text{vol}_{X'}(D') = \text{vol}_X(D)$.*

Proof. It suffices to treat integral D . Projection formula gives

$$H^0(X', \mathcal{O}_{X'}(mD')) = H^0(X, \nu_* \mathcal{O}_{X'} \otimes \mathcal{O}_X(mD)).$$

The quotient η in

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \nu_* \mathcal{O}_{X'} \longrightarrow \eta \longrightarrow 0$$

is supported in dimension at most $n - 1$. Therefore

$$\begin{aligned} h^0(X, \mathcal{O}_X(mD)) &\leq h^0(X', \mathcal{O}_{X'}(mD')) \\ &\leq h^0(X, \mathcal{O}_X(mD)) + h^0(X, \eta \otimes \mathcal{O}_X(mD)) \\ &= h^0(X, \mathcal{O}_X(mD)) + O(m^{n-1}). \end{aligned}$$

Dividing by $m^n/n!$ and taking limsup proves the assertion. $\square$

Theorem 3.2.37 (Continuity of volume). *Let X be a projective variety of dimension $n \geq 1$, and fix a norm $\|\cdot\|$ on $N^1(X)_{\mathbb{R}}$. There is $C > 0$ such that*

$$|\text{vol}(\xi) - \text{vol}(\xi')| \leq C(\max\{\|\xi\|, \|\xi'\|\})^{n-1} \|\xi - \xi'\|$$

for all $\xi, \xi' \in N^1(X)_{\mathbb{Q}}$. The case $\xi = \xi' = 0$ is interpreted as the trivial equality.

Proof. Both sides are homogeneous of degree n , so multiply both classes by a common large integer. Fix very ample Cartier divisors $A_1, \dots, A_r$ whose classes form a basis of $N^1(X)_{\mathbb{Q}}$. By numerical invariance it suffices to treat

$$D = \sum_i a_i A_i, \quad D' = \sum_i a'_i A_i, \quad a_i, a'_i \in \mathbb{Z}.$$

All norms on a finite-dimensional space are equivalent, so use $\|D\| = \max_i |a_i|$. The following argument applies to arbitrary D, D' ; no bigness assumption is required.

First suppose $a_i - a'_i = b_i \geq 0$, and write $D' = D - B$ with $B = \sum_i b_i A_i \geq 0$. Choose very general $E_j \in |A_j|$ meeting all the chosen A_i properly, and put $D_j = D|_{E_j}$. We claim

$$\text{vol}(D - B) \geq \text{vol}(D) - n \sum_{j=1}^r b_j \text{vol}_{E_j}(D_j).$$

For the case $B = b_1 A_1$, choose mb_1 distinct very general divisors $F_\alpha \in |A_1|$ as in the proof of Proposition 3.2.28. Then

$$0 \longrightarrow \mathcal{O}_X\left(mD - \sum_{\alpha=1}^{mb_1} F_\alpha\right) \longrightarrow \mathcal{O}_X(mD) \longrightarrow \bigoplus_{\alpha=1}^{mb_1} \mathcal{O}_{F_\alpha}(mD)$$

is exact at the first two terms. Lemma 3.2.29 identifies every section dimension on the right with $h^0(E_1, mD_1)$. Thus

$$h^0(X, m(D - b_1 A_1)) \geq h^0(X, mD) - mb_1 h^0(E_1, mD_1).$$

Divide by $m^n/n!$, use a subsequence realizing $\text{vol}(D)$, and bound the other term by its limsup. This proves

$$\text{vol}(D - b_1 A_1) \geq \text{vol}(D) - nb_1 \text{vol}_{E_1}(D_1).$$

This argument uses the definition of volume; it does not assert an asymptotic Riemann–Roch equality for the possibly non-nef D_1 .

For several A_j , induct on the number of non-zero b_j . If $\widetilde{D} = D - \sum_{j < r} b_j A_j$, the one-term case gives

$$\begin{aligned} \text{vol}(\widetilde{D} - b_r A_r) &\geq \text{vol}(\widetilde{D}) - nb_r \text{vol}_{E_r}(\widetilde{D}|_{E_r}) \\ &\geq \text{vol}(D) - n \sum_{j < r} b_j \text{vol}_{E_j}(D_j) - nb_r \text{vol}_{E_r}(D_r). \end{aligned}$$

Here restriction of $\sum_{j < r} b_j A_j$ to E_r is effective, so multiplication of sections gives $\text{vol}_{E_r}(\widetilde{D}|_{E_r}) \leq \text{vol}_{E_r}(D_r)$. We may impose the finitely many required very general conditions simultaneously.

Now put $D^+ = \sum_i |a_i| A_i$. The divisor $D^+ - D$ restricts effectively to E_j , so $\text{vol}_{E_j}(D_j) \leq \text{vol}_{E_j}(D^+|_{E_j})$. The latter restriction is nef. Hence

$$\begin{aligned} \text{vol}_{E_j}(D^+|_{E_j}) &= ((D^+)^{n-1} \cdot A_j)_X \\ &= \left(\left(\sum_i |a_i| A_i \right)^{n-1} \cdot A_j \right)_X \leq C_j \|D\|^{n-1}, \end{aligned}$$

where C_j depends only on the fixed very ample basis. Since $\sum_j b_j \leq r\|B\|$, choose $C = 2nr \max_j C_j$ to obtain

$$0 \leq \text{vol}(D) - \text{vol}(D - B) \leq \frac{C}{2} \|D\|^{n-1} \|B\|.$$

For the general case write $D' = D + E - F$, where E, F are non-negative integral combinations of the A_i with disjoint sets of non-zero coefficients. Apply the preceding inequality to (D, F) and (D', E) :

$$\begin{aligned} 0 \leq \text{vol}(D) - \text{vol}(D - F) &\leq \frac{C}{2} \|D\|^{n-1} \|F\|, \\ 0 \leq \text{vol}(D') - \text{vol}(D' - E) &\leq \frac{C}{2} \|D'\|^{n-1} \|E\|. \end{aligned}$$

Since $D - F = D' - E$ and $\max\{\|E\|, \|F\|\} = \|E - F\| = \|D' - D\|$, the triangle inequality gives

$$|\text{vol}(D) - \text{vol}(D')| \leq C \max\{\|D\|, \|D'\|\}^{n-1} \|D - D'\|.$$

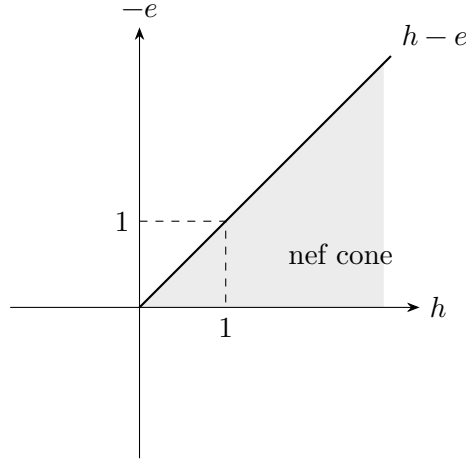
Homogeneity and equivalence of norms finish the proof. $\square$

Corollary 3.2.38 (Volume of real classes). *The function $\xi \mapsto \text{vol}(\xi)$ on $N^1(X)_{\mathbb{Q}}$ extends uniquely to a continuous function $\text{vol} : N^1(X)_{\mathbb{R}} \rightarrow \mathbb{R}_{\geq 0}$.*

Proof. For a rational sequence $\xi_i \rightarrow \xi$, the preceding estimate makes $\text{vol}(\xi_i)$ a Cauchy sequence, since the norms $\|\xi_i\|$ are bounded. Applying the same estimate to two approximating sequences proves that the limit is independent of the sequence. The estimate also proves continuity, and density of rational classes proves uniqueness. $\square$

Example 3.2.39 (Volume on the blow-up of projective space). Let $X = \text{Bl}_P \mathbb{P}^n$, with $n \geq 2$, and let $e, h \in N^1(X)_{\mathbb{R}} \cong \mathbb{R}^2$ be the classes of the exceptional divisor E and the pullback H of a hyperplane. Write a class as $xh - ye$. The nef cone is generated by h and $h - e$, so its region in these coordinates is $x \geq y \geq 0$. In the nef region, volume is the top self-intersection:

$$\text{vol}(xh - ye) = x^n - y^n \quad (x \geq y \geq 0).$$


 Figure 3.1: Plane of $N^1(X)_{\mathbb{R}}$, with coordinates $xh - ye$.

Indeed, $(H^n) = 1$, $(E^n) = (-1)^{n-1}$, and $(H^{n-j} \cdot E^j) = 0$ for $1 \leq j < n$ by the projection formula. On the other hand, for integers $k, l \geq 0$, lE is the fixed part of $|kH + lE|$. If $\pi : X \rightarrow \mathbb{P}^n$ is the blow-up, then

$$\begin{aligned} H^0(X, \mathcal{O}_X(kH + lE)) &= H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(k) \otimes \pi_* \mathcal{O}_X(lE)) \\ &= H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(k)), \end{aligned}$$

because $\pi_* \mathcal{O}_X(lE) = \mathcal{O}_{\mathbb{P}^n}$ by Hartogs across the point P . This is where $n \geq 2$ is needed. Hence in the region $x \geq 0$, $y \leq 0$ spanned by h, e , volume is x^n . Outside these two regions it is zero: for rational classes this follows from the description by polynomials, which have non-negative degree and cannot vanish at P to order greater than their degree. Extend by continuity. Altogether,

$$\text{vol}(xh - ye) = \begin{cases} x^n - y^n, & x \geq y \geq 0, \\ x^n, & x \geq 0, y \leq 0, \\ 0, & \text{otherwise.} \end{cases}$$

The formulas agree on the overlapping boundary $y = 0$.

By rational approximation and continuity, we obtain the following results.

Example 3.2.40 (Volume of a difference of nef Cartier divisors, II). For real nef classes ξ, η on an n -dimensional projective variety,

$$\text{vol}(\xi - \eta) \geq (\xi^n) - n(\xi^{n-1} \cdot \eta).$$

To approximate within nef classes, first add a small positive ample class to both, then approximate the resulting ample classes by rational ample classes. Apply Example 3.2.33 and use continuity of volume and intersection products.

Example 3.2.41 (Volume increases in effective directions). If $\xi \in N^1(X)_{\mathbb{R}}$ is big and $e \in N^1(X)_{\mathbb{R}}$ is effective, then $\text{vol}(\xi) \leq \text{vol}(\xi + e)$. For rational divisors, this follows by multiplying sections by the defining section of a positive integral multiple of the effective divisor. Approximate its non-negative coefficients and those of ξ by rational numbers, then use continuity.

Example 3.2.42 (Birational invariance of volume, III). If $\nu : X' \rightarrow X$ is a birational morphism of projective varieties, then $\text{vol}_X(\xi) = \text{vol}_{X'}(\nu^*\xi)$ for every $\xi \in N^1(X)_{\mathbb{R}}$. This follows from Proposition 3.2.36 by rational approximation and continuity of volume and of the linear pullback map.

3.3 Bigness in Relative Version

In this section we consider bigness for a morphism. By way of preparation, let $f : X \rightarrow S$ be a proper surjective morphism from a normal variety to a projective variety, and assume its geometric generic fibre is irreducible. Write $\eta \in S$ for the generic point, $K = k(\eta)$ and $\overline{K}$ for an algebraic closure. Proper varieties can be treated by pulling line bundles back to projective birational models; the growth argument justifying this extension is given in Lemma 3.3.11 below.

Definition 3.3.1 (Relative Iitaka dimension). The relative Iitaka dimension of a line bundle L on X is

$$\kappa(X/S, L) = \kappa(X_{\overline{\eta}}, L|_{X_{\overline{\eta}}}), \quad X_{\overline{\eta}} = X \times_S \text{Spec } \overline{K}.$$

We keep $\kappa(X, L)$ for the absolute Iitaka dimension.

Remark 3.3.2. If we drop irreducibility of the geometric generic fibre, it is still a normal reduced scheme in our characteristic-zero setting, but may have several disjoint components. Indeed, X_{η} is normal: on affine charts its rings are localizations of the normal rings of X . The extension $\overline{K}/K$ is separable, so normality is preserved by this field extension [SP25, Tag 0C3M]. For $\bar{x}$ lying over $x \in X_{\eta}$, the local description is

$$\mathcal{O}_{X_{\overline{\eta}}, \bar{x}} \cong (\mathcal{O}_{X_{\eta}, x} \otimes_K \overline{K})_{\mathfrak{q}_{\bar{x}}}.$$

Since a noetherian normal scheme is a finite disjoint union of normal integral schemes [SP25, Tag 033M], $X_{\overline{\eta}}$ is a disjoint union of normal varieties. Under our irreducibility assumption it is itself a normal variety.

Definition 3.3.3 (Relative bigness). A line bundle L on a normal variety X is big relative to f , or f -big, if $\kappa(X/S, L) = \dim X_{\overline{\eta}}$, that is, if its restriction to the geometric generic fibre is big. A Cartier divisor D is f -big if $\mathcal{O}_X(D)$ is so.

Remark 3.3.4 (Equivalent definition of relative bigness). In [KMM87] relative bigness is defined using the generic fibre X_{η} . This is equivalent to the geometric generic-fibre definition. Flat base change [Har77, III, 9.3] gives

$$H^0(X_{\eta}, L^{\otimes m}|_{X_{\eta}}) \otimes_K \overline{K} \cong H^0(X_{\overline{\eta}}, L^{\otimes m}|_{X_{\overline{\eta}}}).$$

Thus the dimensions of these spaces over their respective ground fields are the same, and so are the Iitaka dimensions and the bigness conditions. We use either description as convenient. The same definitions extend to non-normal varieties by normalization, as in the absolute case.

Lemma 3.3.5. *Let $f : X \rightarrow S$ be a projective surjective morphism of varieties, with $\dim X_\eta = n$; normality is not required. A line bundle L is f -big if and only if there is $C > 0$ such that*

$$\text{rank } f_* L^{\otimes m} \geq C m^n \quad \text{for every sufficiently large } m.$$

Proof. At the generic point,

$$\text{rank } f_* L^{\otimes m} = \dim_K (f_* L^{\otimes m})_\eta = h^0(X_\eta, L^{\otimes m}|_{X_\eta}).$$

Apply the growth criterion of Lemma 3.2.4, with the all-large-exponents consequence of Corollary 3.2.10. The same proof works over K , or after extension to $\overline{K}$ by the preceding remark. This proves both directions. Surjectivity matters: otherwise the direct image has generic rank zero on S . $\square$

Proposition 3.3.6 (Relative Kodaira's lemma). *Let $f : X \rightarrow S$ be as in Lemma 3.3.5, D an f -big Cartier divisor and F an effective Cartier divisor on X . Then*

$$f_* \mathcal{O}_X(mD - F) \neq 0 \quad (m \gg 0).$$

Proof. The generic fibre of this sheaf is $H^0(X_\eta, \mathcal{O}_X(mD - F)|_{X_\eta})$, which is non-zero for $m \gg 0$ by Kodaira's lemma on the generic fibre. The restriction of F is effective, possibly zero. Thus the direct image is non-zero. $\square$

Corollary 3.3.7 (Characterization of relatively big Cartier divisors). *Let $f : X \rightarrow S$ be a projective surjective morphism of varieties, with S projective, and let D be Cartier on X . The following are equivalent:*

- (i) D is f -big.
- (ii) For every f -ample Cartier divisor A , there are $m > 0$ and an effective Cartier divisor N on X such that $mD \sim_S A + N$.
- (iii) The same statement holds for some f -ample Cartier divisor A .
- (iv) There are an f -ample Cartier divisor A , $m > 0$, and an effective Cartier divisor N with $mD \equiv_S A + N$.

Here $\sim_S$ and $\equiv_S$ have the relative meanings of Appendix A. In (iii) one may also arrange $\sim_{\text{lin}}$ by absorbing a pullback from S into A , which preserves f -ampleness.

Proof. “(i) $\Rightarrow$ (ii)”: Fix A f -ample. For $r \gg 0$, twisting rA and $(r+1)A$ by suitable pullbacks of sufficiently ample divisors on S makes them globally generated. Choose effective representatives

$$rA \sim_S H_r, \quad (r+1)A \sim_S H_{r+1}.$$

Relative Kodaira’s lemma gives a non-zero coherent sheaf $f_*\mathcal{O}_X(mD - H_{r+1})$ for $m \gg 0$. Twist it by a sufficiently ample $\mathcal{O}_S(B)$ to obtain a non-zero global section. By projection formula,

$$\begin{aligned} 0 &\neq H^0(S, f_*\mathcal{O}_X(mD - H_{r+1}) \otimes \mathcal{O}_S(B)) \\ &= H^0(X, \mathcal{O}_X(mD - H_{r+1} + f^*B)). \end{aligned}$$

Thus $mD + f^*B \sim_{\text{lin}} H_{r+1} + N'$ with $N' \geq 0$, and, relatively,

$$mD \sim_S A + (H_r + N').$$

Take $N = H_r + N'$. The implications (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) are immediate. For (iv) $\Rightarrow$ (i), $mD - N$ is relatively numerically equivalent to an f -ample divisor, and hence is f -ample by the fibrewise Nakai–Moishezon criterion. Its restriction to the generic fibre is ample; adding the effective $N|_{X_\eta}$ shows that $D|_{X_\eta}$ is big. $\square$

Corollary 3.3.8. *Under these hypotheses, bigness of a Cartier divisor relative to f depends only on its relative numerical equivalence class.*

Remark 3.3.9. Since the rank is computed on the generic fibre, one may likewise define relative volume by

$$\text{vol}_{X/S}(L) = \limsup_m \frac{\text{rank } f_*L^{\otimes m}}{m^n/n!} = \text{vol}_{X_\eta}(L|_{X_\eta}), \quad n = \dim X_\eta.$$

Real relative bigness and the corresponding generic-fibre cones can be treated similarly. These constructions must be read on the generic fibre, or through restriction to it; they should not be identified without explanation with all of $N^1(X/S)$, which can contain non-zero vertical classes restricting to zero on the generic fibre.

Remark 3.3.10. The proof of Lemma 3.2.4 uses the $O(m^n)$ cohomology estimate for projective schemes, where a Cartier divisor can be written as a difference of very ample divisors. To extend the growth criterion to proper varieties, it suffices to prove the following proper version, as needed for [KM98, 3.22]. The definitions via a projective birational model then agree with the intrinsic growth criterion. On the other hand, a proper morphism with an f -ample line bundle is projective in the convention of Remark 1.5.2; thus the relative ample decomposition in Corollary 3.3.7 naturally retains its projective hypothesis.

Lemma 3.3.11 (Growth of cohomology for proper schemes). *Let X be a proper scheme of dimension n , D a Cartier divisor and $\mathcal{F}$ a coherent sheaf on X . For every $i \geq 0$,*

$$h^i(X, \mathcal{F} \otimes \mathcal{O}_X(mD)) = O(m^n) \quad (m \rightarrow \infty).$$

Proof. We induct on dimension. In dimension zero, X is finite over the ground field, higher cohomology vanishes, and tensoring by a line bundle preserves length, so the assertion holds. A sheaf supported in smaller dimension can be viewed as a sheaf on the closed subscheme defined by its annihilator, to which the induction hypothesis applies.

Step 1: By Chow's lemma choose a projective scheme X' of dimension n and a proper morphism $f : X' \rightarrow X$ which is an isomorphism over an open set U meeting every irreducible component of X . One may first choose a dense quasi-projective such U and apply the version of Chow's lemma preserving it. For the natural map $\mathcal{F} \rightarrow f_* f^* \mathcal{F}$, set

$$\mathcal{K} = \ker(\mathcal{F} \rightarrow f_* f^* \mathcal{F}), \quad \mathcal{J} = \operatorname{im}(\mathcal{F} \rightarrow f_* f^* \mathcal{F}), \quad \mathcal{Q} = \operatorname{coker}(\mathcal{F} \rightarrow f_* f^* \mathcal{F}).$$

Both $\mathcal{K}$ and $\mathcal{Q}$ are supported outside U , in dimension at most $n - 1$. The two exact sequences

$$0 \rightarrow \mathcal{K} \rightarrow \mathcal{F} \rightarrow \mathcal{J} \rightarrow 0, \quad 0 \rightarrow \mathcal{J} \rightarrow f_* f^* \mathcal{F} \rightarrow \mathcal{Q} \rightarrow 0$$

remain exact after tensoring by $\mathcal{O}_X(mD)$. Their long exact cohomology sequences and induction give

$$\begin{aligned} |h^i(X, \mathcal{F}(mD)) - h^i(X, \mathcal{J}(mD))| &\leq h^i(X, \mathcal{K}(mD)) + h^{i+1}(X, \mathcal{K}(mD)), \\ |h^i(X, (f_* f^* \mathcal{F})(mD)) - h^i(X, \mathcal{J}(mD))| &\leq h^{i-1}(X, \mathcal{Q}(mD)) + h^i(X, \mathcal{Q}(mD)). \end{aligned}$$

Here $h^{-1} = 0$. Thus

$$h^i(X, \mathcal{F}(mD)) = h^i(X, (f_* f^* \mathcal{F})(mD)) + O(m^{n-1}).$$

It suffices to bound the right-hand cohomology group.

Step 2: Put $\mathcal{G}_m = f^* \mathcal{F} \otimes f^* \mathcal{O}_X(mD)$. The projection formula identifies $f_* \mathcal{G}_m = (f_* f^* \mathcal{F}) \otimes \mathcal{O}_X(mD)$. Consider the Leray spectral sequence (Appendix D):

$$E_2^{p,q}(m) = H^p(X, R^q f_* \mathcal{G}_m) \implies H^{p+q}(X', \mathcal{G}_m).$$

For each total degree i it induces a decreasing filtration

$$0 = \Phi^{i+1} H^i \subseteq \Phi^i H^i \subseteq \dots \subseteq \Phi^0 H^i = H^i(X', \mathcal{G}_m),$$

with exact sequences

$$0 \longrightarrow \Phi^{p+1} H^i \longrightarrow \Phi^p H^i \longrightarrow E_\infty^{p,i-p}(m) \longrightarrow 0.$$

Taking dimensions and telescoping gives

$$\begin{aligned} h^i(X', \mathcal{G}_m) &= \sum_{p=0}^i (\dim \Phi^p H^i - \dim \Phi^{p+1} H^i) \\ &= \sum_{p+q=i} \dim E_{\infty}^{p,q}(m). \end{aligned}$$

For $q > 0$, the fixed coherent sheaf $R^q f_*(f^* \mathcal{F})$ is supported outside U . The higher-direct-image projection formula gives

$$R^q f_* \mathcal{G}_m \cong R^q f_*(f^* \mathcal{F}) \otimes \mathcal{O}_X(mD).$$

Induction therefore yields $\dim E_2^{p,q}(m) = O(m^{n-1})$ for $q > 0$. Every later term is a subquotient of $E_2^{p,q}$, so

$$h^i(X', \mathcal{G}_m) = \dim E_{\infty}^{i,0}(m) + O(m^{n-1}). \quad (3.2)$$

In the bottom row there is no outgoing differential for $r \geq 2$, since its target has second index $1 - r < 0$. Hence

$$E_{r+1}^{i,0}(m) = E_r^{i,0}(m) / \text{im}(d_r : E_r^{i-r, r-1}(m) \rightarrow E_r^{i,0}(m)).$$

Each image has dimension at most $\dim E_2^{i-r, r-1}(m) = O(m^{n-1})$. There are only finitely many possible incoming differentials, uniformly in m : once $r > i$, the first index of the source is negative. Therefore

$$\dim E_{\infty}^{i,0}(m) = \dim E_2^{i,0}(m) + O(m^{n-1}).$$

Combining this with (3.2) gives

$$h^i(X, f_* \mathcal{G}_m) = h^i(X', \mathcal{G}_m) + O(m^{n-1}).$$

Since X' is projective, the projective cohomology growth estimate applies to the fixed sheaf $f^* \mathcal{F}$ and the fixed divisor $f^* D$, so $h^i(X', \mathcal{G}_m) = O(m^n)$. Step 1 completes the proof. $\square$

The same quotient-sheaf argument as in Proposition 3.2.36 now proves birational invariance of volume, and hence of bigness, for proper varieties. For real Cartier divisors on a proper variety, retain the definition by positive real combinations of big integral Cartier divisors. This is equivalent to bigness after pullback to any projective birational model. Indeed, the integral and rational statements follow from the growth comparison just proved. In the finite-dimensional real span of the finitely many Cartier divisors involved, the inverse image of the big cone on the projective model is open and convex. A real point in it is a convex combination of nearby rational points still in it. Their integral multiples are big on the original variety by the integral statement. This proves the real statement and also its independence of the chosen model.

Example 3.3.12. Let X be a proper variety and N a nef and big Cartier divisor. If T is a nef real Cartier divisor, then $N + T$ is nef and big.

Proof. By Chow's lemma choose a projective variety X' and a proper birational morphism $f : X' \rightarrow X$. The preceding discussion makes f^*N big, and it is nef by pullback of nefness. By Example 3.2.18, for a fixed effective Cartier divisor E and $k \gg 0$ we can write

$$f^*N = A_k + \frac{1}{k}E,$$

where A_k is an ample *rational* Cartier divisor. Since f^*T is nef, $f^*T + A_k$ is ample. Thus

$$f^*(N + T) = (f^*T + A_k) + \frac{1}{k}E$$

is big, and it is also nef. Birational invariance of real bigness gives bigness of $N + T$ on X ; its nefness also follows directly from that of N and T . $\square$

There is a bigness analogue of Proposition 1.5.8.

Proposition 3.3.13. *Let $f : X \rightarrow S$ be a projective surjective morphism of varieties onto a proper variety S . Let D be Cartier on X and M an absolutely big Cartier divisor on S . Then D is f -big if and only if $D + mf^*M$ is absolutely big for all $m \gg 0$.*

Proof. First assume S projective, so X is also projective. “ $\Rightarrow$ ”: By Corollary 3.3.7(iii), absorbing a base pullback into the relatively ample part, write

$$rD \sim_{\text{lin}} A + N, \quad r > 0, \quad A \text{ } f\text{-ample}, \quad N \geq 0.$$

Since M is big on S , there are $q > 0$, an ample divisor H and an effective divisor E with $qM \sim_{\text{lin}} H + E$. Then

$$\begin{aligned} rq(D + mf^*M) &\sim_{\text{lin}} qA + qN + rmf^*(H + E) \\ &= \underbrace{qA + rmf^*H}_{P_m} + \underbrace{qN + rmf^*E}_{G_m}. \end{aligned}$$

The divisor P_m is ample for $m \gg 0$ by Proposition 1.5.8, while $G_m \geq 0$ because f is surjective. Kodaira's characterization therefore shows that $D + mf^*M$ is big. “ $\Leftarrow$ ”: If $D + mf^*M$ is big for even one integer m , there are $t > 0$, an ample Cartier divisor A and $E \geq 0$ with

$$t(D + mf^*M) \sim_{\text{lin}} A + E.$$

Thus $tD \sim_S A + E$, where A is also f -ample. Corollary 3.3.7 gives relative bigness of D .

For a merely proper base S , choose a projective birational modification $\sigma : S' \rightarrow S$ by Chow's lemma, and let X' be the reduced main component of $X \times_S S'$ dominating X :

$$\begin{array}{ccc} X' & \xrightarrow{\tau} & X \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{\sigma} & S \end{array}$$

The variety X' is projective, since f' is projective and S' is projective. The maps σ, τ are birational, and the generic fibre is unchanged. Thus D is f -big if and only if τ^*D is f' -big, and σ^*M is big by proper birational invariance. Apply the projective-base case to (f', τ^*D, σ^*M) , and use

$$\tau^*(D + mf^*M) = \tau^*D + m(f')^*\sigma^*M.$$

Proper birational invariance of bigness transfers the conclusion back to X , in both directions. This proves the stated proper-base version. $\square$

3.4 Examples

3.4A Zariski's Construction

3.4B Cutkosky's Construction

3.4C Okounkov's Construction

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